

The Interplay of Paralinguistic Cues and Expressions of Certainty in Human-Human and Human-Agent Interactions

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There is limited understanding of how paralinguistic cues relate to expressions of certainty in human-agent vs. human-human interactions. We collected data through two phases: (1) we analyzed the pronunciation of phrases from participants across different interaction scenarios and with different certainty levels to establish what cues people use to express certainty; and (2) through an online survey, we presented participants with recordings of humans and agents uttering phrases across different scenarios, and asked participants to identify the certainty level of each recording. We found that it is easier for people to express and recognize low certainty consistently, vs. high certainty, in both human-human and human-agent interactions. Additionally, people tend to rate utterances from agents with lower certainty than humans, even when using the same cues. We contribute design recommendations for agents using paralinguistic cues to express varying levels of certainty across different interaction contexts.

CCS Concepts: • **Human-centered computing** → **HCI theory, concepts and models**; **HCI design and evaluation methods**; **Natural language interfaces**.

Additional Key Words and Phrases: conversational agents, natural language interface, chatbots, virtual assistants, user perceptions, anthropomorphized perceptions, disfluency, fillers, paralinguistic cues, trust calibration

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1 INTRODUCTION

In human-human communication, auditory paralinguistic cues (i.e., para-verbal cues) are known to play many different roles: from conveying emotional states [114, 127], to incorporating cultural norms [46, 118] and expressing intimacy between speakers [88, 121], to conveying different levels of certainty in communication [44, 48, 62]. Unsurprisingly, alongside the growing adoption of conversational interfaces, we also see a growing research interest in understanding how para-verbal cues can be integrated into a conversational agent’s speech [9, 30] to support more complex conversational interactions. For instance, as trust calibration becomes an increasingly important aspect of AI systems design [12, 108, 129], recent studies started exploring ways to incorporate paralinguistic cues for trust calibration [30, 60], highlighting the advantages of expressing uncertainty through ecologically valid and appropriate conversational cues [9]. The interest in affecting trust dynamics through conversational interfaces [66, 117] is further motivated by their high effectiveness in building user trust, e.g. trust in recommendation systems compared to conventional web interfaces [47], even if the presented information is incorrect [26].

Designing trust calibration through the communication of levels of uncertainty [131] requires an ability to control the precision of the communicated information [125], i.e. differentiating levels, which is currently done visually (e.g. graphs [58]), numerically [59, 96, 131], or through textual formats [14, 18, 101]). Indeed, there is evidence that predefined and ordered categorization of uncertainty — levels — is beneficial due to the reduced cognitive effort needed for comprehension [6, 12]. While higher precision methods can be helpful in understanding the underlying workings of the model [12], they require numerical literacy for interpretation and may be difficult for non-domain experts to understand [63]. However, research shows that, for most users, sufficient trust calibration can be achieved even through low-precision methods, such as acknowledging the possibility of being wrong [11, 12, 86]. Thus, although paralinguistic cues can be expected to communicate uncertainty with a lower precision compared to other formats (e.g., numerical or visual), their intuitiveness [9] opens exciting opportunities for trust-calibration applications.

In this work, we explore how voice-based agents can use paralinguistic cues to express different levels of confidence in their statements across different communication contexts. Achieving this understanding can help designers build voice-based agents that more effectively support trust calibration through efficiently communicating certainty across different interaction scenarios and contexts. Little is known on the specifics of various paralinguistic cues associated with the expression of uncertainty, and how these cues are perceived differently between human-human and human-agent interactions. The main findings come from human-human communication research, such as works showing that decreased loudness of voice [115], longer pauses [44], and the use of conversational fillers [72] increase the perception of the speaker’s uncertainty. However, it remains unknown whether and how these cues can be used to discriminate uncertainty levels, i.e., whether humans can reliably recognize different degrees of certainty based on these paralinguistic cues. The matter is further complicated by the evidence that the perception of social cues might differ between human-human and human-agent communications [20, 29], making the direct integration of human communication patterns into agent speech even more challenging.

Our goal was to, first, identify **(RQ1) what paralinguistic cues people use to conversationally express different levels of certainty**, and then, **(RQ2) whether people can differentiate the corresponding levels of certainty expressed by the identified paralinguistic cues in (RQ2.1) human-human and (RQ2.2) human-agent conversations**. Furthermore, to account for the interaction scenario, we considered various forms of machine feedback in

AI-human interaction [25]. In this work, we focus on three interaction scenarios for which the support of natural language interfaces has been previously explored: *item recommendation* [3, 47, 57, 120], *action recommendation* [57, 80, 132], and *object recognition* [68, 74, 85, 132].

The data were collected through two phases of research: (I) a moderated study with follow-up semi-structured interviews (n=30) and (II) an unmoderated online experiment (n=100). In Phase I, we identified seven paralinguistic cues that are used to express different levels of certainty: *volume*, *pace*, *emphasis*, *disfluency*, *pauses*, *pronunciation*, and *intonation (or pitch)*. Specifically, we found that when analyzing the correlation of individual cues on their own to different levels of confidence, *high* confidence is strongly associated with the use of louder volume, faster pace, emphasizing specific words, no-to-minimal prolonged words, and intonation falling at the end of a sentence, while *low* confidence is typically expressed with slower pace, no-to-minimal emphasis of words, some prolongation of words, and rising intonation at the end of the sentence. Most participants had difficulties systematically expressing *medium* certainty, as they tend to think of the expression of certainty as a binary selection (either certain or uncertain). Additionally, and more importantly, we found that there are multiple combinations and variations of these paralinguistic cues that can be used to express low and high certainty, and these combinations also differ across scenario types; thus illustrating that there are multiple ways that these paralinguistic cues can be combined to convey the same level of confidence. While this is the case, there are more variations in how people express *high* certainty compared to low certainty, suggesting that there is less agreement on how to express high certainty compared to low certainty.

Phase II of our study showed that people can consistently recognize low-certainty phrases but had trouble distinguishing phrases between high and medium certainty. Our Phase II findings also revealed that phrases spoken by an agent are consistently rated lower in certainty than phrases with the same paralinguistic cues spoken by a human, thus suggesting that people tend to rate voice-based agents as less certain in their statements than human speakers, even when employing the same paralinguistic cues.

This work contributes an understanding of how paralinguistic cues are used and combined to express different levels of certainty across different interaction scenarios, as well as an understanding of the differences in how these expressions are interpreted differently in human-agent interactions vs. human-human interactions. From this understanding, we propose a set of design guidelines for voice-based agents expressing varying degrees of certainty using paralinguistic cues across different interaction contexts, and discuss ethical considerations of their use.

2 RELATED WORK

Previous research on calibrating user trust in AI systems encompasses various strategies, with a significant focus on communicating the system’s level of certainty. Many of these approaches, however, require domain-specific knowledge, making them less accessible to general users. There is limited research on the use of natural language to express uncertainty, and even less on leveraging paralinguistic cues in human-AI interactions. Additionally, studies suggest that directly applying human-human paralinguistic cues to human-agent conversations is not straightforward. This section reviews the existing literature on how AI agents’ communicate their levels of certainty to users, and the promising application of paralinguistic cues in human-AI conversation, highlighting the gaps our work aims to address.

2.1 AI Agents’ Communication of Levels of Certainty

To achieve effective collaboration between humans and AI systems, a human user needs to be able to establish appropriate levels of reliance on AI system capabilities. This process is typically referred to as *trust calibration* — the alignment between a system’s capabilities and a user’s perceived level of trustworthiness of the system [49, 52, 78]. The

potential for adverse outcomes, particularly in terms of safety, efficiency, and performance, increases when there is a misalignment between humans’ perceived level of system trustworthiness and the actual capabilities of the system, i.e., miscalibrated trust [78, 94, 95]. For example, when a user over-estimates or over-trusts a system’s capabilities, their inflated expectation of system capabilities might result in their acceptance of AI recommendations without critical assessment of recommendation reliability in relation to a given decision task [70, 95, 98]. For instance, Klingbeil et al. [70] found that simply knowing that advice was generated by an AI system caused individuals to overly depend on it, even when it conflicted with available contextual information or their own judgment. Their findings speak to how overreliance frequently results in inefficient outcomes for the advisees themselves, and can also have negative repercussions for third parties. On the other hand, when a user under-estimates or under-trusts a system, they may resist adopting or using AI tools that could potentially enhance or streamline their decision-making tasks or processes. For example, an area of focus in the space of intelligent personal assistants (e.g., Siri, Alexa, Google) has been the tension between the benefits of personalized assistance and users’ privacy concerns [19, 83, 110]. In both scenarios — overreliance and underreliance — miscalibrated trust leads to sub-optimal utilization of the AI system, and impedes the collaborative efficiency between humans and AI.

One common approach to mitigating the negative effects of miscalibrated trust in human-AI interactions involves AI systems clearly communicating their levels of certainty to users. By articulating their level of confidence or certainty, the AI system enables users to assess the reliability of the recommendations generated by the AI system and thus appropriately adjust their level of reliance on the system [49, 52, 78]. There exist various strategies currently implemented to enable the communication of an AI system’s level of certainty to the human interacting with the system, employing different formats, such as visual aids (e.g., Visual Map [58]), numerical values (e.g., probability scores or confidence intervals/percentages [59, 96, 131]), or text-based explanations [14, 18, 101]), to present the information to the user. Different strategies to represent certainty in AI systems can be thought of as existing along a spectrum. For example, Bhatt et al. [12] characterize this spectrum along the trait of precision. At one end, certainty is conveyed with more detailed and specific information—a ‘high-precision communication method’. At the other end, certainty is communicated in a more broader and generalized manner—a ‘low-precision communication method’.

The selection of the most suitable strategy, in terms of the format and level of precision with which a representation of certainty is expressed, requires careful consideration of the communication context, and in particular, the communication recipient [12, 33, 106, 107]. For instance, in high-stakes environments like healthcare [55, 84] or autonomous driving [4, 5, 27], or in situations where the end-user interacting with an AI system is a domain expert [8, 119], more detailed and precise communication of certainty is necessary to ensure safety and effectiveness. In the case of numerical expressions of uncertainty, a level of numerical literacy is necessary among communication recipients to meaningfully calibrate trust to appropriate levels [11, 12]. Conversely, in more routine or less critical applications, such as at-home interactions with an intelligent personal assistant [93, 109], simpler and lower-precision indications of certainty might suffice and even be preferable to avoid inundating the user with too much information [11, 12, 86]. With the widespread adoption of voice-based intelligent personal assistants¹, there is a need for widely accessible ways through which such agents can express their levels of certainty to users; a need for more intuitive, easier-to-understand indicators. There is an opportunity for such voice-based agents to use voice not only for communicating their messages but also as a medium for non-verbally expressing their levels of confidence through para-verbal cues.

¹<https://www.statista.com/statistics/973815/worldwide-digital-voice-assistant-in-use/>

2.2 Paralinguistic Cues in Human-Human Conversations

In human-to-human conversations, one of the ways people naturally convey their level of certainty is through the use of paralinguistic cues [7, 115]. These para-verbal cues of communication, which include tone of voice, pitch, rate of speech, rhythm, and stress on particular words, complement and enhance the meaning of the spoken message [7, 36, 115]. Paralinguistic cues represent a dimension of interpersonal speech that allows individuals to indicate their level of confidence to their conversation partner(s), especially in situations of uncertainty and ambiguity [7, 115]. For example, a rising intonation at the end of a sentence can indicate uncertainty, while a steady and confident tone can convey assurance [7, 115]. In such interactions, linguistic and paralinguistic elements interact almost instinctively, providing additional context and meaning to message recipients. The organic integration of these cues helps listeners pick up on subtleties that words alone might not convey. For instance, a speaker's hesitation, changes in volume, or emphasis on certain words can reveal their true feelings, intentions, or levels of certainty [28, 51]. These cues enable a deeper understanding of the speaker's message, allowing listeners to interpret not just what is being said, but also how it is being said and the underlying emotions or doubts [7, 115]. This nuanced communication fosters clearer understanding, reduces the risk of misinterpretation, and supports more effective decision-making to occur within the context of the conversation and beyond.

However, the expression and interpretation of paralinguistic cues can be affected by differences in demographics such as cultural background, language proficiency, and individual characteristics. Studies have shown that people from different backgrounds infer the emotional state of their communication partners modulated by their cultural knowledge and social rules [15, 82, 130]. For example, the non-verbal cue of silence is associated with communication that has gone wrong (i.e., communication breakdowns) in Western cultures such as the United States [126], whereas in Eastern cultures such as Japan, silence can convey deep thinking and respect [32]. These cultural differences in paralinguistic communication can be partly explained by the theory of linguistic relativity, also known as the *Sapir-Whorf Hypothesis*. This theory suggests that the structure of a language influences how its speakers perceive, interpret, and categorize the world [71, 111, 112, 128]. Since language and culture are deeply intertwined, speakers of different languages may learn to express and understand paralinguistic cues in culturally-specific ways. As a result, individuals have an in-group advantage to correctly identify the context associated with paralinguistic cues in their native linguistic and cultural frameworks [75, 99, 130].

Similar to cultural background, language proficiency also shapes how interlocutors use and interpret paralinguistic cues [100]. There is evidence that higher language proficiency is correlated with better accuracy and sensitivity to intonation [16]. Mennen [87] found that German speakers of English exhibit a narrower pitch range than native English speakers, which may be perceived as more negative. Also, [45] found major differences between native and non-native listeners in their ability to identify emotions expressed in voice. Collectively, these findings indicate that greater language proficiency facilitates more accurate and nuanced interpretation of paralinguistic cues during communication.

In addition to cultural and language proficiency differences, individual characteristics also affect the interpretation of paralinguistic cues. For example, Casillas et al. [16] found that individuals with higher empathy are more sensitive to intonation cues to form sound-meaning associations. Also, the gender of the listener can influence the perception of the communication partner, such as their perceived personality traits [10]. Lee and Boster [77] found that slower speech rate increased the perception of credibility of Korean male speakers, while rapid delivery is associated with higher credibility for Korean female speakers.

While demographic differences affect the perception of paralinguistic cues, research shows that paralinguistic cues can be reliably perceived by participants. Participants from different backgrounds (Western vs. remote Namibian villages) were able to universally recognize basic emotions from non-verbal communication across cultural boundaries [75, 99, 113]. In both groups (speakers from Austria and Germany) significantly lower word rate and lower speaking rate as well as higher rates of filled pauses and hesitation markers are found in formal (polite) conditions [46]. In Semnani-Azad and Adair [118] study of vocal cues between East versus the West, they found several vocal cues with universal and shared meanings across both cultures. Hence, it is worthwhile to explore leveraging paralinguistic cues in human-AI dialogue, with attention paid to the various cultural and linguistic factors discussed above that may affect the perception of these cues.

2.3 Leveraging Paralinguistic Cues in Human-AI Dialogue

In tandem with the rapidly developing capabilities of conversational agents, boosted by large language models, the research community is increasingly inclined to explore the use of paralinguistic cues as a novel, more naturalistic way to represent and present voice-based AI agents' levels of certainty [9, 30]. To illustrate, Becker et al. [9] argues that 'incompetence markers' or 'markers of error' such as the disfluencies (a type of paralinguistic cue) that occur in natural human speech aid in shaping appropriate expectations in listeners about their dialogue partner. In the current state of conversational interfaces, the absence of these markers, the authors argue, leads users to interpret this as their absence of incompetence as well, creating a gulf in expectations [9]. Similarly, Dubiel et al. [30] agree that paralinguistic cues such as disfluencies and elongations, can be leveraged to indicate an AI agent's uncertainty, and that this approach is relevant to fostering trust calibration.

While human-human conversations can inspire more naturalistic interactions in human-AI dialogues, research shows that the principles and dynamics of human-human interactions cannot be directly and exactly transferred and operationalized in the design of human-agent communication. Both Doyle et al. [29] and Clark et al. [20] highlight that current conversational agents lack the social and relational dynamics of human conversations. While human-like qualities are important for creating relatable agents—those that users can understand and find engaging—merely *emulating* human conversation falls short of user expectations and the functional purpose of these agents. They argue for designing conversational agents with their own distinct norms and expectations, focused on utilitarian aspects rather than full replication, better addressing the unique dynamics and goals of human-AI interactions. Thus, while human-human conversation serves as a valuable model for guidance, direct application to human-AI conversation must be adjusted to account for these differences.

In the context of uncertainty expression in conversational agent design, this distinction becomes even more critical. For example, Kim et al. [67] found that expressing uncertainty through natural language, particularly using first-person phrases, decreases user confidence and reliance on AI responses while improving accuracy. However, these first-person uncertainty expressions can also reduce users' desire to trust and use the AI system. Jeong et al. [61] further found that conversational fillers like "um" and "uh" make voice-based conversational agents seem less intelligent and likable in task-oriented interactions, but more entertaining in social conversations. While fillers did not significantly affect the perceived human-likeness of conversational agents, their appropriateness varied by context, suggesting designers should avoid fillers in precise, task-focused scenarios but consider them for enhancing engagement in social contexts. This underscores the necessity for tailored strategies in human-AI dialogue design, where the unique dynamics of uncertainty expression and conversational nuances must be carefully balanced to maintain both accuracy and user

trust. Therefore, while drawing inspiration from human-human interaction, it is essential to adapt these principles to meet the specific needs and contexts of human-AI communication.

In summary, while researchers have started to explore the use of paralinguistic cues by voice-based agents to express different levels of certainty, we cannot simply apply knowledge about how humans use paralinguistic cues for expressing certainty to the design of voice-based agents for the same expressions. Furthermore, designers do not have a clear set of design guidelines for which cues to use to express varying degrees of certainty across different interaction tasks and contexts, or an understanding of how paralinguistic cues used for expressions of certainty are perceived differently when coming from agents vs. when coming from humans.

Therefore, in this work, we explore what paralinguistic cues people use to express different levels of certainty, and whether people can differentiate these levels in both human-human and human-agent conversations. Voice is a rich source of information [92], conveying nuances through prosodic features (e.g., tone, pitch, pacing), which significantly affect how verbal messages are perceived. Understanding these cues can potentially enhance the design of conversational agents, making them more effective at calibrating user trust. By examining these cues in both human-human and human-agent interactions, we aim to bridge the gap between human communication practices and AI interactions, addressing the unique dynamics of each to ensure conversational agents manage user expectations appropriately and effectively.

3 PHASE I: EXPRESSING DIFFERENT LEVELS OF CERTAINTY

In order to understand how voice-based agents can use and combine paralinguistic cues to convey varying levels of certainty across different interaction contexts, it helps to first understand how people use such cues to conversationally express different levels of certainty (RQ1). To explore this, we conducted a moderated user study. Specifically, we collected and analyzed the pronunciation of nine phrases across three interaction scenarios (*item recommendation*, *action recommendation*, and *object recognition*), each with three certainty levels (*low*, *medium*, and *high*) from 30 participants. We also conducted semi-structured interviews with each participant to gain insight into their self-reported strategies for certainty expression. The study protocol was approved by our institution's research ethics board.

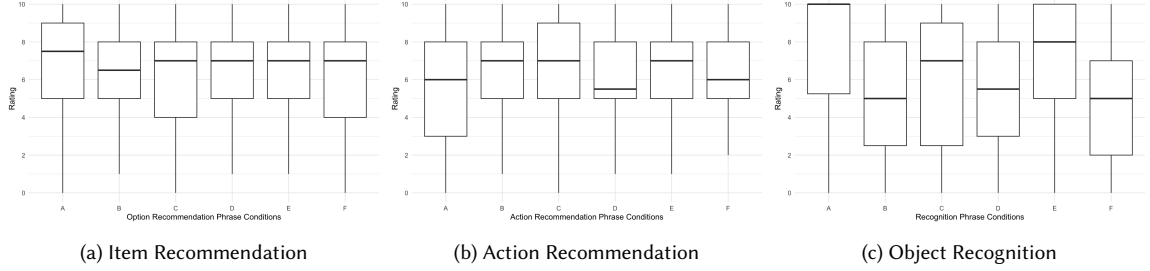
3.1 Study Design and Data Analysis

To support low, yet sufficient precision ordered categorization of uncertainty [6, 12] for potential trust calibration purposes, we followed certainty categorization used by Hyland and Milton [56] and Radensky et al. [105] and used three levels of certainty — high, medium, and low — to be expressed by participants for the study. This is in line with the three levels of epistemic modality individuals use to assess the likelihood that something is true or not as defined in human-human communication literature [54, 81]. To account for the potential effects of an interaction scenario, we considered various forms of machine feedback in AI-human interaction Dellermann et al. [25] and focused on three scenarios commonly supported through natural language interfaces: *item recommendation* [3, 47, 57, 120], *action recommendation* [57, 80, 132], and *object recognition* [68, 74, 85, 132]. For each scenario, we then needed to identify stimuli phrases to be pronounced by participants with different levels of certainty to allow us to capture the variations in paralinguistic cues. To enhance the internal validity in both Phase I and the following Phase II, we wanted to control for the certainty conveyed solely by the content of each phrase, ensuring the between-phrase similarity of confidence levels for clear comparability in analysis. Thus, to develop our stimuli set, we first conducted a pre-study.

Table 1. The phrases per scenario used for pre-study.

	Item Recommendation (IR)	Action Recommendation (AR)	Object Recognition (OR)
Phrase A	I would recommend this option.	You should take this action.	This is a cat.
Phrase B	I would suggest going with this option.	I would advise this course of action.	What we have here is a cat.
Phrase C	This would be the choice I'd recommend.	This is the course of action I'd advise.	The object is a cat.
Phrase D	My recommendation would be this option.	I suggest you move forward with this action.	This is recognized as a cat.
Phrase E	This is the option I would suggest.	This action should be your next step.	The picture is showing a cat.
Phrase F	This would be the choice I'd recommend.	You should move forward with this action.	This is deemed to be a cat.

Fig. 1. Box plots for phrase conditions per scenario.



3.1.1 Pre-Study to Select Phrases. For each of the three interaction scenarios, we intended to have three pre-determined phrases that conveyed comparable confidence for participants to express different levels of certainty, thus capturing the variations in the use of paralinguistic cues. We began by generating six phrases with similar meanings for each scenario (Table 1). These phrases were then incorporated into a survey as 18 questions asking users to provide a rating for the level of certainty expressed by each phrase from zero (not confident at all) to ten (very confident). The survey was configured to randomly display nine out of the 18 phrases, and the order of questions was randomized to avoid ordering bias.

A priori one-way ANOVA power analysis [40] was conducted using G*Power software [35] to estimate the minimum sample size needed for this study. While this model assumes parametric distribution of the data and does not account for random effects of the variation across participants, the estimation provides a fair approximation for the number of data points needed. To achieve a statistical power of 80% and a medium effect size Cohen's $f = 0.25$ [21] (also based on comparable HCI studies [2, 41, 103]) with 0.05 significance comparing across 6 different phrases, we need a minimum of 216 data points for each scenario, or an average of 36 ratings per phrase. Based on this, we decided to recruit a minimum of 100 participants, thus giving us on average 50 ratings per phrase.

Through a crowdsourcing platform², we recruited 109 participants from the United States who were fluent in English and consented to partake in the survey. The participants are 46% male / 54% female, with ages ranging from 18 to over 60 years old. Out of these participants, 86 (79%) provided completed responses, 23 (21%) provided partial responses. This presented us with on average 45 ratings per phrase, with 276 ratings in the item recommendation scenario, 264 ratings in the action recommendation scenario, and 279 ratings in the object recognition scenario.

After collecting the ratings for each phrase in each scenario (visualized in Figure 1), we used the linear mixed-effect model [73] to perform a pairwise comparison between phrases for each scenario (Table 2) to test for statistical

²<https://www.surveymonkey.com/market-research/solutions/audience-panel/>

Table 2. The p-values of pairwise comparisons between phrases using the linear mixed-effect model for each scenario. Green highlighted cells showcases the three selected phrases per scenario that have the highest combined p-value.

<i>Phrase Comparison</i>	Scenarios		
	<i>Item Recommendation</i>	<i>Action Recommendation</i>	<i>Object Recognition</i>
A - B	0.6301	0.6366	0.0001
A - C	0.9045	0.7579	0.0231
A - D	0.9525	1.0000	<.0001
A - E	0.6085	0.7208	0.7446
A - F	0.0412	0.9961	<.0001
B - C	0.9895	0.9997	0.4878
B - D	0.9641	0.6568	0.9908
B - E	1.0000	0.9999	0.0115
B - F	0.7133	0.8647	0.6611
C - D	1.0000	0.8793	0.1856
C - E	0.9943	1.0000	0.5495
C - F	0.3871	0.9599	0.0084
D - E	0.9803	0.8049	0.0003
D - F	0.2308	0.9995	0.9352
E - F	0.7551	0.9604	<.0001

Table 3. The selected phrases per scenario and their corresponding ratings.

Item Recommendation		Action Recommendation		Object Recognition	
Phrase	Rating	Phrase	Rating	Phrase	Rating
(B) I would suggest going with this option.	6.31	(B) I would advise this course of action.	6.57	(B) What we have here is a cat.	5.30
(C) This would be the choice I'd recommend.	6.51	(C) This is the course of action I'd advise.	6.45	(D) This is recognized as a cat.	5.04
(E) This is the option I would suggest.	6.34	(E) This action should be your next step.	6.47	(F) This is deemed to be a cat.	4.65

significance and select phrases that are most similar to each other. As a result, we selected three phrases per scenario (listed in Table 3) with the highest combined pairwise-comparison p-values. Lastly, we ran a one-way ANOVA test [40] to confirm the ratings for our chosen phrases are not significantly different from each other:

- $\Pr(>F) = 0.847$ for Item Recommendation Phrases B, C, E
- $\Pr(>F) = 0.747$ for Action Recommendation Phrases B, C, E
- $\Pr(>F) = 0.551$ for Object Recognition Phrases B, D, F

3.1.2 Phase I Main Study Design. Next, to establish what paralinguistic cues people use to express low, medium, and high levels of certainty for the selected phrases, we designed a moderated study followed by a semi-structured interview.

We recruited 30 participants through university bulletin boards and online social platforms. Before the study session, we collected demographic information for each participant. The participants were geographically situated in North America, 18 to over 70 years old (56% participants in the 18-29 range), 50% self-identified women and 50% self-identified men. As part of our recruitment criteria, all participants were fluent in English, with 53% of them who are native English speakers and 47% who are not.

We invited each participant to an hour-long study session on Zoom. During the study session, following initial instructions and a brief warm-up exercise to familiarize participants with the format of the study, we presented participants with three scenarios — item recommendation, action recommendation, and object recognition — asking them to pronounce each of the three pre-determined phrases (Table 3) for each scenario (presented one-by-one) with low,

Table 4. Average Value for Each Paralinguistic Cue

<i>Paralinguistic Cue</i>	<i>Measurement</i>	Level of Certainty		
		<i>High</i>	<i>Medium</i>	<i>Low</i>
Volume	% difference in RMS from medium level	5.15%	0%	-1.01%
Pace	% difference in clip length from medium level	-13.49%	0%	23.07%
Emphasis	Number of emphasized words used in phrase	0.77	0.10	0.08
Pauses	Number of pauses used in phrase	0.49	0.57	1.36
Disfluency	Number of fillers / repetitions used in phrase	0.04	0.11	0.47
Prolonged Words	Number of prolonged words used per phrase	0.12	0.38	1.10
Ending intonation - rising	% of total phrases ending in rising intonation	1.85%	20.00%	54.81%

This table outlines the average value for the paralinguistic cues at each level of certainty, as well as the measurement method used.

medium, and high levels of certainty. The order in which to present the scenarios, the phrases, and the level of certainty were randomized to eliminate ordering bias. Each phrase was recorded individually as an audio file on the interviewer’s computer through the Zoom session. Participants had the opportunity to listen to each phrase and re-record until they are satisfied with their utterances. As a result, we collected 27 utterance pronunciations per participant (3 scenarios $\times$ 3 phrases $\times$ 3 levels of certainty). We then conducted a semi-structured interview, focused on probing participants’ considerations influencing their choice of paralinguistic cues for expressing different levels of certainty.

To prepare all 810 collected utterance pronunciations (30 participants $\times$ 27 expressions per participant) for the quantitative analysis, we first trimmed each audio clip to remove silences at the beginning and the end, then measured the length of the utterance in seconds, and the perceived volume based on root mean square (RMS) of the audio track. We observed absolute volume variations across participants based on their environment during the study, as well as their equipment for recording. To normalize volume across utterances for comparison, we instead measured the relative volume differences for high and low levels of certainty as compared to the same phrase expressed for medium level of certainty (e.g. the high level of confidence utterance is 5% louder than the medium level of confidence for the same phrase by the same participant). Similarly, we normalized pace as the relative difference in audio clip length for high and low levels of certainty as compared to the same phrase expressed for medium level of certainty (e.g. the high level of confidence utterance is 10% faster than the medium level of confidence for the same phrase by the same participant).

We then applied conversation analysis to each utterance by using a modified version of the Jeffersonian Transcription System [102] (some notations are modified to differentiate features in data analysis). Specifically, we denoted words that are emphasized (**word**), noticeable pauses (*(.)*), words that are prolonged (*wo:rd*), ending in rising intonation (*?*), and ending in falling intonation (*_*). We also preserved disfluencies, such as conversational fillers (e.g. *um*, *uh*) and repetitions in the transcription. Here is a sample transcribed phrase: *Um *this* (.) is deemed to be (.) a cat_*. For each utterance, we had one member of the research team transcribe the audio, and a second member to cross-check the transcriptions. Disagreements in transcription were brought before the whole research team to discuss and revise until the team reached a consensus. For each utterance, these transcriptions were then analyzed to calculate the number of occurrences for the following paralinguistic cue — emphasis, disfluencies (fillers and repetitions), pauses, and prolonged words. As for analyzing the use of ending intonation, since each utterance was coded as either ending in rising or falling intonation, we calculated the cue ‘ending intonation - rising’ as the percentage occurrence across all utterances and used it in our analysis (e.g. across all high certainty utterances, 2% of them ended with rising intonations / 98% of them ended with falling intonations).

To analyze the use of these paralinguistic cues across all 810 utterances, we first generated a summary statistics table showcasing their average values for each level of certainty. We then ran a multinomial logistic regression model with random effects [34] on our data across the paralinguistic cues to identify the combinations that are statistically significant in predicting high and low levels of certainty using medium level of certainty as the baseline. In the model, we chose participant ID as the random effect. Variance Inflation Factor (VIF) [37] was used to ensure the model had no significant multicollinearity problems with all variables having a VIF of no more than 4 [1]. Next, we applied pairwise comparison using the linear mixed-effect model [73] for each level of certainty and paralinguistic cue, with English native speaking as the fixed effect and participant ID as the random effect. We also performed clustering analysis to partition the utterances into unique archetypes, helping us to understand different combinations of paralinguistic cues that participants used to express different levels of certainty. To do this, we first used the gap statistic [123] to determine the optimal number of clusters based on our data, then applied the k-means clustering algorithm [50] to assign each data point to a specific cluster. Finally, we repeated the statistical analyses above, multinomial logistic regression with random effects and clustering analysis, for each scenario and level of certainty to investigate if the difference in scenarios affected participants' expressions.

For the qualitative analysis, we fully transcribed audio recordings for each session, then thematically analyzed the data using a deductive approach based on Gale et al. [38]'s Framework Method. We used three themes in our codebook: paralinguistic cues, levels of certainty, and scenarios. Within the theme of paralinguistic cues, the codes are based on cues analyzed in the quantitative analysis above: *volume*, *pace*, *emphasis*, *disfluency*, *pauses*, *prolonged words*, and *ending intonation*. In addition, we also included *enunciation* capturing quotes that are related to the pronunciation of words, as well as *pitch* as literature suggests these cues are used to expression of uncertainty [69, 97]. We used codes *high*, *medium*, *low* for levels of certainty, and *item recommendation*, *action recommendation*, *object recognition* for scenarios based on our study design. Two researchers first performed independent coding for three participant transcripts using codes from the predetermined codebook and recording any new codes that were discovered. The researchers then discussed and compared their codes, agreeing the codebook was sufficient for analysis and could be applied to all subsequent transcripts. The researchers then analyzed the rest of the participant transcripts independently, cross-checked each other's analyzed data, and resolved discrepancies through multiple rounds of discussions until agreement was reached. Following this consensus, the extracted codes and quotes were systematically charted into an analytical framework matrix, which the whole research team then used to collectively interpret the data.

3.2 Findings

We identified seven paralinguistic cues that people use to express different levels of certainty: *volume*, *pace*, *emphasis*, *disfluency*, *pauses*, *pronunciation*, and *intonation (or pitch)*. Our multinomial logistic regression results reveal that, when looking at each paralinguistic cue individually, *high* confidence is strongly associated with the individual cues of *louder volume*, *faster pace*, *emphasizing specific words*, *no-to-minimal prolonged words*, and *intonation falling at the end of a sentence*; while *low* confidence is strongly associated with the individual cues of *slower pace*, *no-to-minimal emphasis of words*, *some prolongation of words*, and *rising intonation at the end of the sentence*. However, when looking at different variations of how these paralinguistic cues are used and combined together, our clustering results reveal that there are multiple combinations and variations of these cues that people use to express low and high confidence, thus revealing that there are multiple ways that people can use and combine these cues to express low and high certainty. These combinations and variations also vary across scenario types (item recommendation, action recommendation, and object recognition); however, our clustering results also revealed that, overall, **there are more ways that people use and**

Table 5. Summary of the paralinguistic cues associated with different levels of certainty

Paralinguistic Cue	Level of Certainty		
	High (context dependent)	Medium	Low (more universal)
Volume	louder (~5%)	neutral baseline	softer (~1%)
Pace	faster (~13%)		slower (~23%)
Emphasis	keywords (~1 per phrase)		no-to-minimal
Pauses	no-to-minimal, before emphasis		~1 per phrase
Disfluency	none		may use in casual situations
Prolonged words	no-to-minimal		~1 word per phrase
Ending Intonation	falling		rising
Enunciation	concise, clear		mumbling, stuttering, trailing off
Pitch	lower		higher, fluctuations

Green highlighted cells denote the cues that are statistically significant to differentiate from medium certainty baseline.

combine these cues to express high certainty than there are for low certainty, which suggests that there is less agreement among participants on how to express high certainty compared to low certainty. This finding was backed up by participants' interview responses, which suggest that people are more consistent in how they express low certainty compared to high certainty.

3.2.1 Medium Level of Certainty.

From our findings, medium level of certainty does not have a defining set of paralinguistic cues corresponding to it, and it is instead often associated with minimal paralinguistic cues and a neutral way of speaking that is in-between how participants would convey high and low levels of certainty.

Almost all participants associated medium level of certainty with being neutral:

"...medium, it was more just neutral." [P36]; "...medium, one, I was going for more of a neutral tone." [P80]

Expressing medium level of certainty in a neutral way was also viewed as a natural baseline:

"(medium) sounded most like how you would normally speak like your natural baseline" [P23]; "When I talk medium confidence, I just try to be as natural as possible" [P92]

Participants perceived the neutrality of medium level of certainty as trying not to influence their listeners to act in any particular way as they did not feel strongly about their statements:

"...medium to me is like I'm not biasing you either way. I'm just telling you what it is" [P18]; "...for medium, I would go for the most neutral way to express something .. I don't feel very strongly about either way. I don't care if you take this option or not." [P47]

To express medium level of certainty as being neutral, participants tried to minimize the paralinguistic cues used in their expressions to communicate the idea of nothingness:

"...with medium .. this gives me the idea of nothingness" [P07]; "(medium) I wasn't trying to communicate anything" [P18]

As such, they were purposefully keeping their expressions plain and machine-like:

"medium, I'd probably just say it plainly" [P72]; "...for medium, just reading the sentence like a robot or something" [P23]

Also, participants expressed the phrases similar to how they would normally speak in everyday in relaxed situations:

Table 6. High Level of Certainty: Summary Statistics & Multinomial Regression Results

<i>Paralinguistic Cue</i>	<i>Summary Statistics</i>		<i>Multinomial Regression</i>	
	<i>average</i>	<i>stdev</i>	<i>coefficient</i>	<i>p-value</i>
Volume	5.15%	0.05	9.72	<0.0001
Pace	-13.49%	0.46	-1.98	0.0003
Emphasis	0.77	0.67	3.77	<0.0001
Pauses	0.49	0.63	-0.30	0.1894
Disfluency	0.04	0.25	-0.73	0.1116
Prolonged Words	0.12	0.34	-1.08	0.0007
Ending Intonation - Rising	1.85%	0.14	-3.66	<0.0001

The summary statistics columns showcases the mean values and standard deviations of each paralinguistic cue. The multinomial logistic regression columns presents the p-values and coefficient estimates of each cue in the multinomial logistic regression model representing *high* level of certainty relative to *medium* level of certainty. Green cells shows cues with p-values that are statistically significant ($p < 0.0500$).

"I kind of describe (medium) as just my everyday speak just being as neutral as possible" [P35]; "...medium is is just a relaxed way of talking" [P92]

Since participants did not have any defining paralinguistic cues associated with medium level of certainty, they modified their expressions relative to high and low levels of certainty to stay neutral in the middle:

*"I don't think I had a clear strategy for medium other than just making it relative to high and low confidence" [P90];
"...for medium it was probably the hardest because you kind of had to like go between both of them (high and low)" [P35]*

Given that medium level of certainty is perceived as neutral, we decided to use medium certainty as the baseline for our statistical tests.

3.2.2 High Level of Certainty.

When looking at the combination of studied paralinguistic cues to distinguish high level of certainty utterances from medium level, the expression of high certainty is associated with the use of *louder volume*, *faster pace*, *emphasizing on key words*, the *lack of prolonged words*, and *ending phrases with a falling intonation*. However, our clustering analysis illustrates that there are multiple ways for people to express high certainty using paralinguistic cues. These differences may be based on the conversation situation (e.g. delivering a presentation or talking to a friend), or the perceived personal stake of the interaction scenario (e.g. recognizing an object or making a recommendation).

Running our data of 810 utterances through the multinomial logistic regression model with random effects, we found that the following combination of paralinguistic cues were statistically significant in predicting high level of certainty as compared to medium level of certainty: *louder volume*, *faster pace*, *emphasizing on keywords*, *the lack of prolonging words*, and *ending on a falling intonation* (Table 6). The model did not find the use of pauses or disfluencies to be significant in differentiating between high and medium levels of certainty. Each of these paralinguistic cues is discussed in more detail below (statistically significant cues are denoted with *).

Volume*. We found that participants were 5.2% louder in volume when expressing the phrases in high level of certainty compared to medium level (Table 4). Based on the results of our multinomial regression, the coefficient for volume is statistically significant, with the multinomial log-odds of the level of certainty being high relative to medium level of certainty increasing by at least 9.7 times when volume increases (Table 6).

During the interview sessions, 22 out of 30 participants brought up volume variations as a self-reported strategy for distinguishing between levels of certainty. Specifically, participants noted that they were using higher volume (speaking louder) to convey persuasiveness in high certainty expressions:

"...for high confidence I am trying to persuade you by using louder voice." [P23]; "Also I increase the volume a bit so I show that I am really confident about it." [P92].

Pace*. Participants were 13.5% faster in their pace of speech when expressing the phrases in high level of certainty as compared to medium level of certainty (Table 4). Based on the results of our multinomial regression, the coefficient for pace is statistically significant, with the multinomial log-odds of high level of certainty decreasing by at least 2.0 times when the pace of speech increases (slower pace) (Table 6).

When analyzing the self-reported strategies brought up by participants in relation to their pace of speech, we found that 19 out of the 30 participants mentioned using speed as a variant to express different levels of certainty. The majority of participants indeed self-reported that they tried to use faster pace of speech for high certainty levels:

"I speak faster on high than on low" [P07].

Some participants explained that faster pace is associated with high level of certainty because their brains are processing thoughts at a higher speed with minimal hesitation:

"...so when I'm very confident and probably at that time, my brain is also processing thing at a higher speed." [P72]; "...for high, probably a a quicker, non hesitating" [P81].

However, a small number of participants mentioned that expressing high level of certainty could result in slower pace, as they are purposefully enunciating words for clarity:

"...it's to bring clarity so like when you enunciate words, you kind of moving your lips more and saying them so it could be slower" [P18]

Emphasis*. We found that participants emphasized on average 0.77 words per phrase when expressing high certainty, compared to emphasizing less than 0.1 words per phrase when expressing medium or low certainty (Table 4). Based on the results of our multinomial regression, the coefficient for emphasis is statistically significant, with the multinomial log-odds of high level of certainty increasing by at least 3.8 times when the use of emphasized words increases (Table 6).

Our qualitative data supports this finding, as 23 out of 30 participants chose to use emphasis as a paralinguistic cue to distinguish between different levels of certainty. Participants discussed choosing keywords to emphasize higher levels of confidence:

"...(high) they have more emphasis on certain words to get your point across" [P19]; "(using emphasis on high) to make it more of a statement" [P29]

Interestingly, we found that there are different strategies used by participants in selecting the keywords to emphasize in a phrase. The most common heuristic is to emphasize on the object like *"this"*, *"the option"*, or *"the object"*:

"If I'm super confident about it, I'm like 'this', this is what you're gonna do" [P85]; "If I'm high confidence means I really trust that's the decision I made is correct. So I'll emphasize the suggestion, the actual option I suggest." [P65]

The emphasis could also be placed on the verb, like *"suggest"*, *"should"*, *"recognize"*:

"But in high .. I'll maybe emphasize the 'should' part in that one." [P75]

Another common heuristic is to emphasize on “I” in the sentence, attributing confidence towards themselves:

“...if I am sure .. emphasizing like ‘I’ as in this is, you know, my decision .. kind of that like responsibility thing.” [P54];
“...sometimes I would highlight the ‘I’, so I’m suggesting this to you.” [P18]

Participants varied their strategies to emphasize on words, as some mentioned that they could express the phrase in different ways to convey the same level of confidence, or changed the words to emphasize depending on the sentence structure:

“I would choose different words that I was gonna highlight.” [P18]; “...if the keywords are at the beginning or the end, then you can either build it up then bring it down.” [P07]

Pauses. Participants used fewer pauses for high level of certainty with average of 0.49 pauses per phrase as compared to medium level of certainty with average of 0.57 pauses per phrase (Table 4). However, the coefficient for the use of pauses is not statistically significant based on our multinomial regression model to distinguish high level of certainty from medium (Table 6).

In our qualitative analysis, most of the participants did not explicitly mention using pauses as a paralinguistic cue to differentiate their utterances from other levels of certainty, supporting our regression results above. Of the few participants who mentioned the use of pauses for high level of certainty, they were deliberately adding pauses to their speech as a style of presentation.

“...for high level of confidence, I am deliberately pausing even though I know what to say next because I wanna make a point” [P47]; “(high confidence) I do become more conscious of where I’m pausing .. so that it’s almost like a better presentation” [P89].

Disfluency. In our analysis, we classified disfluency as the use of fillers (e.g. “um”, “ah”) and repetitions (e.g. “This is .. this is”). Participants generally refrain from using disfluencies (average 0.04 per phrase) to express high level of certainty (Table 4). Based on the results of our multinomial regression, the coefficient for disfluency is not statistically significant to differentiate between high and medium levels of certainty (Table 6).

Our qualitative analysis showed that participants deliberately chose not to use disfluencies for high level of certainty:

“When you [have] high confidence, you don’t need to go for [fillers] .. you just go for it.” [P18];

Prolonged Words*. We found participants used the least amount of prolonged words (avg. 0.12 per phrase) when expressing high certainty, compared to avg. 0.38 per phrase for medium level of certainty (Table 4). Based on the results of our multinomial regression, the coefficient for emphasis is statistically significant, with the multinomial log-odds of high level of certainty decreasing by at least 1.0 times when the number of prolonged words increases (Table 6).

In our qualitative analysis, we did not find any participants who mentioned the use of prolonged words to express high level of certainty.

Ending Intonation*. For high level of certainty phrases, 2% of these utterances ended with a rising intonation, which means 98% ended with a falling intonation (Table 4). Based on the results of our multinomial regression, the coefficient for ending intonation is statistically significant, with the multinomial log-odds of high level of certainty decreases by at least 3.5 times when the utterance ends in a rising intonation (Table 6).

Table 7. High Level of Certainty: Pairwise Comparison Between Native and Non-native English Speakers

Paralinguistic Cue	Average		Comparison
	Native	Non-Native	<i>p-value</i>
Volume	3.94%	6.53%	0.1461
Pace	-17.41%	-9.01%	0.5453
Emphasis	0.715	0.841	0.4497
Pauses	0.285	0.722	0.0045
Disfluency	0.014	0.071	0.2345
Prolonged words	0.063	0.190	0.0296
Ending Intonation - Up	0.69%	3.17%	0.3112

Pairwise comparison between native and non-native English speakers using linear mixed-effect model for each paralinguistic cue expressed for high level of certainty. Green cell shows the comparisons that are statistically significant at 95% confidence level

Participants' quotes support the finding that falling ending intonation is often used to express high level of certainty:

"...if it's high confidence, I was trying to end it more on the down." [P35]; "[high]...I drag the word down." [P89]

Other Paralinguistic Cues. There were two other paralinguistic cues we found in our qualitative analysis that are associated with expressing high level of certainty: **clear enunciation of words** and **lower pitch of voice**.

Some participants used clear enunciation as a pronunciation strategy to show high confidence:

"Enunciating with clarity is very important [for high confidence]... I'm not mumbling the words." [P18]; "So I show that I'm really confident about it, I'm trying to be like, concise and like specific." [P92]

Lowering pitch of voice was another strategy used to express high level of certainty. Interestingly, this strategy was attributed to public figures who purposefully lowered their voices in order to increase their projected confidence.

"...my voice is probably gonna be on the lower end, more basic, it was constant when I was in high confidence." [P46]; "...but I think the CEO of Theranos would use a deeper voice to show confidence, and people who knew her personally would say, that's not how she usually sounds." [P47]

Linguistic and Cultural Factors

Prior literature suggests that participants from different cultural and language backgrounds may vary in their ways of using paralinguistic cues in their vocal expressions (e.g. [82, 100, 130]). Thus, we analyzed and compared how native English vs. non-native English speaking participants used paralinguistic cues to express levels of certainty (Table 7). Based on our analysis, we noticed there are a couple of cues where the difference between native and non-native English speakers are statistically significant. Specifically, native English speakers on average used less *pauses* (0.285) and *prolonged words* (0.063) compared to non-native English speakers (pauses = 0.722, prolonged words = 0.190). This finding suggests that language proficiency may affect how paralinguistic cues are used to express high level of certainty.

During participant interviews, several participants mentioned mimicking the paralinguistic cues of other people to express high level of confidence. Specifically, some participants mentioned cultural icons such as CEOs and sovereigns, as well as characters from TV shows, are used as their standard for vocal expressions of high confidence. This suggests culture could be an influencing factor in vocal expressions because individuals may consider how others within their culture express high confidence when deciding how to express high confidence themselves.

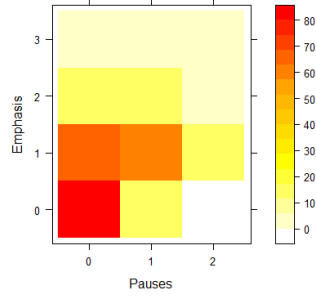
Table 8. High Level of Certainty: K-means Cluster Analysis

	No.	Volume	Pace	Emphasis	Pauses	Disfluency	Prolonged Words	Ending Intonation - Rising
Cluster 1	81	3.30%	13.40%	1.01	1.23	0.07	0.25	2.47%
Cluster 2	72	5.89%	-16.41%	0	0.15	0.01	0.04	0%
Cluster 3	64	4.89%	-19.36%	1.02	0	0.02	0.06	4.69%
Cluster 4	28	4.37%	17.37%	2.11	0.61	0.07	0.21	0%
Cluster 5	25	10.52%	-111.71%	0.12	0.16	0.04	0	0%

The k-means clusters for high level of certainty across all the measured paralinguistic cues.

Fig. 2. High Level of Certainty: Heatmap of Pauses vs. Emphasis

Heatmap of Pauses vs Emphasis in all Scenarios



“Another thing that I did was put emphasis on certain words, and I think that’s a strategy that I think I’ve seen other people use in order to portray high levels of confidence.” [P29]; “typically for the higher confidence ones I’m kind of stating it in like how I picture a king would say a statement.” [P90]; “(for high confidence) mimicking things that I’ve watched, and a lot of that comes from the TV show Friends.” [P46]

Clustering Analysis

Our clustering analysis reveals five unique archetypes of utterances associated with a high level of certainty across all the measured paralinguistic cues as dimensions (Table 8), thus illustrating that there are at least five combinations of these cues from our data that tend to be used to express high certainty. This suggests that there are multiple ways for people to express high certainty, and that people are less consistent in how they use the measured paralinguistic cues to express high confidence.

The paralinguistic cues *pace*, *emphasis* and *pauses* have the biggest variations across the five clusters. Clusters 1 and 4 both include the use of emphasized words and pauses in the high certainty expressions, while these two cues are minimally used by participants in clusters 2 and 5. Cluster 3 shows that participants emphasized on keywords, but did not use any pauses in their expressions.

While pause as a standalone paralinguistic cue is not statistically significant to differentiate between high vs. medium level of certainty, we find in our qualitative analysis that some participants used the strategy of *combining emphasized words with deliberate pauses* to convey high confidence:

Table 9. High Level of Certainty Per Scenario: Summary Statistics & Multinomial Regression Results

<i>Paralinguistic Cue</i>	<i>Item Recommendation (IR)</i>			<i>Action Recommendation (AR)</i>			<i>Object Recognition (OR)</i>		
	average	coefficient	p-value	average	coefficient	p-value	average	coefficient	p-value
Volume	4.71%	13.44	0.0044	5.10%	11.64	0.0020	5.63%	10.59	0.0037
Pace	-6.30%	-1.32	0.2522	-10.66%	-2.26	0.0392	-23.50%	-3.47	0.0015
Emphasis	1.00	3.65	<0.0001	0.74	3.63	<0.0001	0.58	7.99	<0.0001
Pauses	0.40	-1.03	0.0375	0.59	-0.28	0.4859	0.48	0.01	0.9755
Disfluency	0.08	0.83	0.2737	0.03	-0.80	0.2826	0.01	-7.22	0.0980
Prolonged Words	0.10	-1.94	0.0220	0.14	-1.22	0.0178	0.12	-1.04	0.1574
Ending Intonation - Rising	4.44%	-1.97	0.0126	0%	-10.58	0.6436	1.11%	-8.94	0.0002

The summary statistics column (average) showcases the mean of each paralinguistic cue. The multinomial logistic regression columns (coefficient & p-value) presents the p-values and coefficient estimates of each cue in the multinomial logistic regression model representing *high* level of certainty relative to *medium* level of certainty. Green cells shows cues with p-values that are statistically significant ($p < 0.0500$).

"...high confidence, I would emphasize, I would deliberately pause" [P47]; "...high is like more pause, more focused on a few like stronger intonation (on words)" [P18].

Visualizing the relationship between the cues emphasis and pauses as a heatmap shows that indeed the combination of these two cues is being used frequently in the expression of high certainty, alongside of not using any emphasis and pauses, as well as using emphasis only without pauses (Figure 2).

Our qualitative analysis of participants' transcripts also revealed that participants are differentiating their expressions of high certainty *based on the situation or context*, such as delivering a presentation or talking to a friend:

"...when I'm confident, then I speak very fast"; "[for high certainty] I actually may say it slowly, if I'm like like presenting." [P72]; "... particularly when I'm presenting something rather than talking to a friend .. I do become more conscious of where I'm pausing." [P89]

One of the contextual situations driving differences in high certainty expressions is the conversation scenario (item recommendation, action recommendation, object recognition) presented to participants. The cluster (combination of cues) differences are discussed in more detail in the following section.

Scenario Differences

Alongside our analysis of the overall combination of cues for high certainty, we were interested in analyzing how the cues are used together to convey high certainty across the three interaction scenarios: *item recommendation*, *action recommendation*, and *object recognition*. We found that while volume and emphasis are significant across all three scenarios, the significance of the other paralinguistic cues differ based on the scenario. Clustering analysis revealed that within each scenario there are different ways of expressing high certainty, such as the use of emphasis and pauses. Based on our qualitative analysis, it indicates that the perceived personal stake of the conversation may be a factor affecting differences in expressing high level of certainty.

The results from the multinomial logistic regression model with random effects reveal differences of the overall group of cues associated with high certainty across scenarios (Table 9). In all three scenarios, *volume* and *emphasis* are statistically significant in predicting high certainty as compared to medium certainty utterances. However, while the paralinguistic cues *pace*, *prolonged words*, and *ending intonation* are statistically significant overall, their significance vary depending on the interaction scenario: pace of speech is not significant for the item recommendation scenario; the

Table 10. High Level of Certainty Per Scenario: K-means Cluster Analysis

		No.	Volume	Pace	Emphasis	Pauses	Disfluency	Prolonged Words	Ending Intonation - Rising
IR	Cluster 1	23	3.59%	5.17%	1	1.04	0.04	0.04	4.35%
	Cluster 2	15	4.47%	2.88%	1	0	0	0	0%
	Cluster 3	11	5.21%	-53.18%	1	0	0	0	0%
	Cluster 4	10	4.70%	0.14%	0	0	0	0	0%
	Cluster 5	9	5.17%	25.36%	2.22	1.11	0.22	0.33	0%
	Cluster 6	9	8.54%	-44.57%	0	0	0	0	0%
	Cluster 7	7	3.62%	2.26%	2.14	0	0	0.14	0%
	Cluster 8	5	4.42%	5.25%	1	0	0.20	0.60	60%
	Cluster 9	1	-0.64%	-14.93%	1	2	3	1	0%
AR	Cluster 1	38	2.33%	18.57%	1.13	1.26	0.03	0.26	0%
	Cluster 2	30	5.62%	-30.30%	0	0.10	0.03	0.07	0%
	Cluster 3	13	6.03%	-2.06%	1.15	0	0	0.08	0%
	Cluster 4	9	13.69%	-80.99%	1	0.22	0.11	0	0%
OR	Cluster 1	32	6.30%	-16.01%	0	0.31	0	0.06	0%
	Cluster 2	24	2.73%	22.32%	1.17	1.29	0.04	0.29	4.17%
	Cluster 3	20	3.93%	-17.41%	1.20	0	0	0.10	0%
	Cluster 4	12	11.69%	-107.22%	0	0.17	0	0	0%
	Cluster 5	2	10.60%	-251.97%	0	0	0	0	0%

The k-means clusters for *high* level of certainty across all the measured paralinguistic cues as dimensions **across the three different scenarios** (object recognition clusters are labelled with ‘OR’; item recommendation clusters are labelled with ‘IR’; action recommendation clusters are labelled with ‘AR’).

use of prolonged words is not significant for the object recognition scenario; and ending intonation is not significant for the action recommendation scenario. Interestingly, we found that *pauses* is a statistically significant cue in the item recommendation scenario, but not significant overall or for any other scenarios.

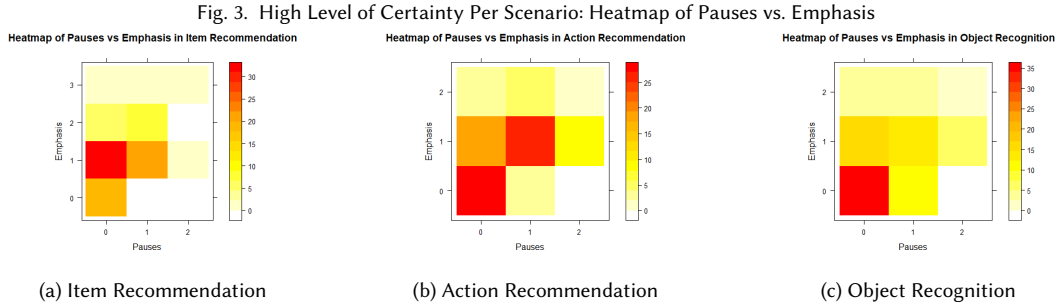
The results from our clustering analysis for high level of certainty across all the measured cues as dimensions, when broken down by each interaction scenario, reveals *nine* clusters in the data for the *item recommendation* scenario, *four* clusters in the data for the *action recommendation* scenario, and *five* clusters in the data for the *object recognition* scenario (Table 10). This suggests that there are different ways people are using the measured paralinguistic cues in expressing high level of certainty within each of the scenarios.

Specifically in the *item recommendation* scenario, the multinomial regression model predicted the following group of statistically significant cues and their characteristics to be associated with high certainty as compared to medium certainty: *louder volume, more emphasized words, fewer pauses, minimal prolongation of words, and ending phrases on a falling intonation* (Table 9). The nine clusters for this data suggests that people use paralinguistic cues quite differently from each other, with the majority of the expression differences coming from *pace of speech, emphasis, and pauses* (Table 10). For example, some participants in this scenario expressed high certainty using a faster pace (-53%), some emphasis on words (1) and no pauses (0) (IR cluster 3), while others used a slower pace (25%), more emphasis on words (2.22), and some pauses (1.11) (IR cluster 5).

In the *action recommendation* scenario, the multinomial regression model predicted the following group of statistically significant cues and their characteristics to be associated with high certainty: *louder volume, faster pace of speech, more emphasized words, and minimal prolongation of words* (Table 9). There are four clusters of paralinguistic cues for the expression of high certainty, suggesting some differences in the usage of these cues but less varied compared to the item recommendation scenario above (Table 10). The majority of the differences comes from *pace of speech, emphasis, and*

pauses. For example, some participants chose to express high level of certainty with slower pace (19%), some emphasized words (1.13) with some pauses (1.26) (AR cluster 1), while others used faster pace (-30%) with no emphasized words (0) and minimal pauses (0.1) (AR cluster 2).

In the *object recognition* scenario, the multinomial regression model predicted the following group of statistically significant cues and their characteristics to be associated with high certainty: *louder volume*, *faster pace of speech*, *more emphasized words*, and *ending with falling intonation* (Table 9). There are five clusters of paralinguistic cues for the expression of high certainty, suggesting some differences in the usage of these cues with similar number of variations as compared with the action recommendation scenario, and less variations compared to the item recommendation scenario (Table 10). The majority of the differences comes from *volume*, *pace of speech*, *emphasis*, and *pauses*. For example, some participants chose to express high level of certainty with slightly higher volume (3%), slower pace (22%), some emphasized words (1.17) with some pauses (1.29) (OR cluster 2), while others used much higher volume (12%), faster pace (-107%) with no emphasized words (0) and minimal pauses (0.17) (OR cluster 4).



Taking a closer look at the relationship between *emphasis* and *pause* for high certainty utterances for each interaction scenario (Figure 3), we found that there are differences in the usage of these two paralinguistic cues. In the *item recommendation* scenario, the most common expression is using zero pauses and some emphasized words, followed by using some emphasis and pauses, as well as using no emphasis and pauses (Figure 3a). There are slight differences in the usage of these cues in the *action recommendation* scenario, where most participants tend to use zero emphasis and zero pauses, or using some emphasis and some pauses (Figure 3b). Lastly, the *object recognition* scenario is quite different from both item recommendation and action recommendation scenarios, as the majority of the participants did not use any emphasis or pauses in their expressions (Figure 3c).

While further research is needed to investigate the reasoning behind the differences in certainty recognition between scenarios, Phase I findings indicated that the *perceived personal stake* of the conversation may be a direction to pursue. For the *object recognition* scenario, participants did not feel like there is a lot of personal stake involved as it is a statement of fact instead of a personal opinion:

"I think the cat one (object recognition) ..., you're you're making a statement, like you're stating a fact." [P36]; " ... (object recognition) if you get it wrong, it's not so bad (as compared to) when you're suggesting an option to somebody." [P79]

For both the *item recommendation* and *action recommendation* scenarios, participants mentioned that they felt like there is more personal stake in making a recommendation for someone else to take:

Table 11. Low Level of Certainty: Summary Statistics & Multinomial Regression Results

<i>Paralinguistic Cue</i>	<i>Summary Statistics</i>		<i>Multinomial Regression</i>	
	<i>average</i>	<i>stdev</i>	<i>coefficient</i>	<i>p-value</i>
Volume	-1.01%	0.09	-1.94	0.2228
Pace	23.07%	0.26	6.34	<0.0001
Emphasis	0.08	0.28	-1.17	0.0018
Pauses	1.36	1.14	0.18	0.2621
Disfluency	0.47	0.83	0.30	0.2493
Prolonged Words	1.10	1.12	0.37	0.0207
Ending Intonation - Rising	54.81%	0.50	1.70	<0.0001

The summary statistics columns showcases the mean values and standard deviations of each paralinguistic cue. The multinomial logistic regression columns presents the p-values and coefficient estimates of each cue in the multinomial logistic regression model representing low level of certainty relative to *medium* level of certainty. Green cells shows cues with p-values that are statistically significant ($p < 0.0500$).

".. with like a recommendation with the high confidence ones, I felt more like I'm commanding someone versus the recognition one." [P90]; "I think the idea of ownership comes into play a bit more over there because a friend is asking for a recommendation." [P80]

Between the two recommendation scenarios, recommending an action was perceived as having higher personal stake as compared to recommending an item:

".. difference between action recommendation versus option (item) recommendation, (it's) more personal." [P18]; "... (for) action recommendation, you're trying to help your friends through by counseling your friend on what to do .. (item recommendation) I was imagine more something a lot more minor like places to go or tools to pick things like that." [P89]

Because of the perceived higher personal stake associated with the action recommendation scenario, participants felt that they needed to be more confident in order to express their opinion:

"(action recommendation) was more like a commanding action .. it was really weird for me to make a recommendation where I didn't feel confident." [P18]; "If I had low confidence I wouldn't probably wouldn't advise somebody (for action recommendation)." [P79]

3.2.3 Low Level of Certainty.

When looking at the combination of studied paralinguistic cues for low certainty compared with medium certainty, we found that the expression of low certainty is associated with phrases spoken in a *slower pace*, containing *fewer emphasized words*, *more prolonged words*, and *ending phrases on a rising intonation*. We also found that people obfuscated their pronunciation of words, as well as using higher or varying pitch of voice when expressing themselves with low confidence. Our clustering analysis illustrates that people are quite consistent expressing low certainty using paralinguistic cues overall as well as across interaction scenarios, with some variations in how pauses, disfluencies, and prolonged words are being used. We also found that pauses, disfluencies and prolonged words may be used interchangeably by participants as a way to buy themselves some thinking time.

When running our data of 810 utterances through the multinomial logistic regression model with random effects, we found that the following combination of paralinguistic cues were statistically significant in predicting low level of certainty relative to medium level of certainty: *slower pace*, *minimal emphasis on words*, *more prolongation of words*, and *ending on a rising intonation* (Table 11). The model did not find volume, the use of disfluencies, or pauses to be

significant in differentiating between low and medium levels of certainty. Each of these paralinguistic cues is discussed in more details below (statistically significant cues are denoted with *).

Volume. Low level of certainty phrases were 1% softer as compared to medium level of certainty (Table 4). Based on the results of our multinomial regression, the coefficient for volume is not statistically significant to differentiate between low and medium levels of certainty (Table 11). However, participants mentioned during interview sessions using softer voice as a strategy to express low level of certainty:

“I might use a lower volume as a proxy for low confidence.” [P47]; “...and when you’re not confident you seem to just not speak up for yourself.” [P50]

Pace*. Participants were 23.1% slower in their pace of speech when expressing the phrases in low level of certainty as compared to medium level of certainty (Table 4). Based on the results of our multinomial regression, the coefficient for pace is statistically significant, with the multinomial log-odds of low level of certainty increasing by at least 6.3 times when the length of utterance increases (slower pace) (Table 11).

Participants mentioned that they are slower in their speech, as a way of showing that they are unsure and still thinking through their responses:

“...low level confidence would mean that the person is still formulating the sentence, making the statement in the middle of them speaking. So that’s why I would think it’s it would lead to slower speech.” [P52]; “...when I when I feel less confident, I’m more slow, like, I’m like taking my time to think about it.” [P34]

Emphasis*. When participants expressed low level of certainty, most of them did not emphasize any words (0.08 per phrase) in their utterances (Table 4). Based on the results of our multinomial regression, the coefficient for emphasis is statistically significant, with the multinomial log-odds of low level of certainty increasing by at least 1.2 times when the number of emphasized words decreases (Table 11).

During interview sessions, participants mentioned consciously not emphasizing words in low certainty situations as a strategy to express reduced personal accountability:

“I’m sort of reducing accountability by kind of, you know, not emphasizing, like almost trying to hide behind it.” [P81]; “...my way of telling if I’m confident or not would be how much emphasis I’m putting on certain words [P16]”

Pauses. Participants used on average 1.36 pauses per phrase when expressing low certainty, compared to about 0.57 pauses per phrase for medium certainty (Table 4). Based on the results of our multinomial regression, the coefficient for pauses is not statistically significant to differentiate between low and medium levels of certainty. (Table 11).

Our qualitative analysis demonstrates that participants are associating with the increased occurrence of pauses with the expression of low certainty:

“What I think I did to show [low] uncertainty level is to add more pauses.” [P46]; “I am purposely making the pause to to give you the notion that I am thinking about what I’m saying and in between the lines, I’m trying to say I am doubting what I’m saying.” [P07]

In addition, participants mentioned taking longer pauses in their low certainty utterances:

“[I take] longer pauses in low confidence.” [P52]; “For low, we would have more pregnant pauses like, I’m telling you this, but I’m not really sure.” [P18]

Disfluency. Comparing across the different levels of certainty, we found that the use of disfluencies – fillers (e.g. um, ah) and repetitions of words – is higher for the expression of low certainty with average of 0.47 disfluencies per phrase as compared to high and medium levels of certainty with fewer than 0.1 disfluencies per phrase (Table 4). Based on the results of our multinomial regression, the coefficient for disfluency is not statistically significant to differentiate between low and medium levels of certainty (Table 11).

During the interview sessions, 16 out of the 30 participants mentioned using fillers or repetitions in their expressions to convey uncertainty. Many participants mentioned using disfluencies for low level of certainty as a way to buy time for themselves to think about what to say:

“I throw filler words in there when I’m trying to think in the moment of what I’m going to say.” [P54]; “I also probably repeat words when I’m low in confidence, or when I’m thinking like what to say.” [P72]

Interestingly, participants noted that they are unconsciously using disfluencies in their speech, therefore they do not intentionally plan on using them as a way to differentiate between levels of certainty:

“[Fillers] do come up in more of a natural conversational format. But when I was reading it off the slides, the ‘ums’ and ‘ahs’ didn’t come to mind.” [P16]; “We usually don’t plan when we use fillers, so it’s more of a unconscious thing, too. Probably that’s why I haven’t used them in the [phrase] recordings, but I did use them in this interview.” [P92]

On the other hand, some participants are consciously minimizing the use of disfluencies from their speech, as their usages are perceived to be unprofessional:

“I have been trained as a public speaker and giving presentations and I’ve had a lot of supervisors who were very adamant that we don’t ever introduce it [fillers] in my speech.” [P52]; “I do speeches professionally so it’s very important to keep track of filler words that I use and try to minimize their usage in my speech.” [P92]

Prolonged Words*. We found that participants used the most amount of prolonged words when expressing low certainty (average = 1.1 per phrase) compared to medium certainty (average = 0.38 per phrase) and high certainty (average = 0.12 per phrase). Based on the results of our multinomial regression, the coefficient for prolonged words is statistically significant, with the multinomial log-odds of low level of certainty increasing by at least 0.4 times when the number of prolonged words increases (Table 11).

Participants’ quotes from our interview sessions supported the finding that the strategy of elongating words is a strategy associated with expressing low level of certainty, as a way to give themselves more time to think:

“...there’s a lot of lagging the words .. the elongation of the the words, almost like a method to keep the sentence formulation going to give themselves more thinking time.” [P52]; “I show uncertainties like elongating more.” [P46]

Ending Intonation*. For low level of certainty, 55% of the phrases ended with a rising intonation (Table 4). Based on the results of our multinomial regression, the coefficient for ending intonation is statistically significant, with the multinomial log-odds of low level of certainty increasing by at least 1.7 times when the utterance ends in a rising intonation (Table 11).

During interview sessions participants discussed ending their phrases on a rising intonation as a strategy to express low certainty as a way to phrase the sentence as a question:

“...kind of like raising your voice at the end of a sentence to show it’s a question kind of thing. I was using that more in low.” [P54]; “(for low certainty) I’m almost raising the last part as a question.” [P47]

Participants wanted their utterances to sound like a question for lower certainty expressions as it leaves the interpretation open for the receiver of the message and showcases the speaker questioning themselves:

“(for low) I kind of finish my statement almost with a little bit of an inflection to make it seem like a question to kind of leave it open and prompt the person I’m talking to.” [P36]; “(in low certainty) I try to speak with a high intonation (at the end of sentence), kind of like second guessing myself.” [P18]

Other Paralinguistic Cues. There were two other paralinguistic cues we found in our qualitative analysis that are associated with expressing low level of certainty: **obfuscating pronunciation of words**, as well as **higher or varying pitch of voice**.

For low certainty, participants wanted to be less clear in their pronunciation of words as a way of reducing their presence in the conversation:

“...for low, I was trying to hide ‘I’ somewhere in there. I did not want it to be the most clear word in the sentence.” [P80]; “...try to make my voice sound like a little bit broken.” [P04]

To be obfuscate their expressions, participants trailed off at the end of sentences, or added stutter in their speech:

“I think the whole thing with low confidence, kind of trailing your sentence off, or leaving it open.” [P36]; “I think I would stutter sometimes.” [P11];

On the other hand, participants mentioned using higher pitch overall to convey low level of certainty:

“...when I am on the uncertainty at several moments I feel that my pitch would go high.” [P46]; “...for low I use a bit more of a high pitch tones... because it’s like the questioning feel, and it’s very unsure.” [P19]

Also, participants mentioned fluctuating their pitch more often when they are low in confidence:

“[For low] there was a bit more fluctuation there, I wanna say, almost like a fluttering.” [P18]; “...and if I’m varying my pitch too much, it probably means I’m not confident in what I’m saying.” [P16]

Linguistic and Cultural Factors

Based on our pairwise comparison between native and non-native English speakers for each paralinguistic cue used to express low level of certainty (Table 12), we did not see any cues that showed a statistically significant difference between the two. This finding suggests that language proficiency may not be an influencing factor in how individuals express themselves in low confidence situations.

In our participant interviews we did not find any mentions of cultural factors influencing their expressions for low certainty. Participants seem to be choosing their vocal cues to convey their internal state of mind, such as adding long pauses or disfluencies allowing themselves more time to think, or ending their statements in an upward intonation to question the validity of their claims (discussed in the paralinguistic cues section above).

Clustering Analysis

Table 12. Low Level of Certainty: Pairwise Comparison Between Native and Non-native English Speakers

Paralinguistic Cue	Average		Comparison
	Native	Non-Native	<i>p-value</i>
<i>Volume</i>	-1.19%	-0.81%	0.8314
<i>Pace</i>	18.40%	28.44%	0.0980
<i>Emphasis</i>	0.063	0.107	0.3998
<i>Pauses</i>	1.146	1.603	0.1355
<i>Disfluency</i>	0.472	0.468	0.9868
<i>Prolonged words</i>	0.924	1.302	0.1428
<i>Ending Intonation - Up</i>	44.44%	66.67%	0.0580

Pairwise comparison between native and non-native English speakers using linear mixed-effects model for each paralinguistic cue expressed for low level of certainty. Green cell shows the comparisons that are statistically significant at 95% confidence level

Table 13. Low Level of Certainty: K-means Cluster Analysis

	No.	Volume	Pace	Emphasis	Pauses	Disfluency	Prolonged Words	Ending Intonation - Rising
Cluster 1	184	-1.29%	14.38%	0.09	0.79	0.16	0.58	57.07%
Cluster 2	86	-0.41%	41.65%	0.06	2.57	1.14	2.22	50.00%

The k-means clusters for low level of certainty across all the measured paralinguistic cues as dimensions across all the scenarios.

Fig. 4. Low Level of Certainty: 3D Scatter Plot of Pauses, Disfluency, and Prolonged Words



The gap statistic analysis results reveal two unique clusters for low level of certainty across the measured paralinguistic cues as dimensions (Table 13), thus illustrating that there are at least two combinations of these cues from our data that tend to be used to express low certainty. This suggests that people are more consistent in how they use the measured paralinguistic cues to express low certainty (2 clusters) as compared to high certainty (5 clusters). Utterances in both clusters tend to be slightly lower volume, slower pace, use minimal-to-no emphasis, and ending with rising intonation.

Table 14. Low Level of Certainty Per Scenario: Summary Statistics & Multinomial Regression Results

<i>Paralinguistic Cue</i>	<i>Item Recommendation (IR)</i>			<i>Action Recommendation (AR)</i>			<i>Object Recognition (OR)</i>		
	average	coefficient	p-value	average	coefficient	p-value	average	coefficient	p-value
Volume	-0.73%	0.04	0.9925	-2.44%	-6.44	0.0414	0.13%	0.99	0.7148
Pace	25.07%	8.62	<0.0001	21.58%	9.11	<0.0001	22.55%	4.06	0.0007
Emphasis	0.12	-1.26	0.0424	0.10	-1.63	0.0132	0.03	-1.13	0.3207
Pauses	1.32	-0.07	0.8009	1.44	-0.13	0.6662	1.31	0.62	0.0287
Disfluency	0.54	0.90	0.0952	0.43	0.04	0.9329	0.43	-0.14	0.7525
Prolonged Words	1.04	0.22	0.4798	1.18	0.74	0.0142	1.08	0.28	0.2907
Ending Intonation - Rising	55.56%	1.50	0.0005	43.33%	1.63	0.0004	65.56%	2.32	<0.0001

The summary statistics column (average) showcases the mean of each paralinguistic cue. The multinomial logistic regression columns (coefficient & p-value) presents the p-values and coefficient estimates of each cue in the multinomial logistic regression model representing *low* level of certainty relative to *medium* level of certainty. Green cells shows cues with p-values that are statistically significant ($p < 0.0500$).

Table 15. Low Level of Certainty Per Scenario: K-means Cluster Analysis

		No.	Volume	Pace	Emphasis	Pauses	Disfluency	Prolonged Words	Ending Intonation - Rising
IR	Cluster 1	90	-0.73%	25.07%	0.12	1.32	0.54	1.04	55.56%
AR	Cluster 1	90	-2.46%	21.58%	0.10	1.44	0.43	1.18	43.33%
OR	Cluster 1	63	0.19%	13.27%	0.02	0.81	0.16	0.49	73.02%
	Cluster 2	27	0.00%	44.22%	0.06	2.48	1.07	2.44	48.15%

The k-means clusters for *low* level of certainty across all the measured paralinguistic cues as dimensions **across the three different scenarios** (*object recognition* clusters are labelled with 'OR'; *item recommendation* clusters are labelled with 'IR'; *action recommendation* clusters are labelled with 'AR').

Comparing between the two clusters, the main differences are the use of *pauses*, *disfluencies* and *prolonged words*. The majority of utterances represented by cluster 1 (184 data points) are using less of these cues as compared to cluster 2 (86 data points).

In our qualitative analysis, we found that participants associated *pauses*, *disfluencies* and *prolonged words* with buying themselves some thinking time to formulate their thoughts as discussed above for each of these paralinguistic cues. Participants also mentioned using these cues, specifically fillers and pauses, interchangeably to express uncertainty:

"I guess the fillers and the pauses are kind of interchangeable." [P47]; "... and sometimes (for low certainty) I did throw in an 'um' or something like that, or a pause." [P99]

Upon visualizing the usage of pauses, disfluencies, and prolonged words in a 3D scatter plot (Figure 4), we see that the data points are uniformly distributed across the graph, suggesting that there are no discernible patterns in how our participants used these cues. This potentially supports our qualitative finding that these cues are used interchangeably to express low confidence.

Scenario Differences

When running the multinomial logistic regression model with random effects on the utterances for the three interaction scenarios (*item recommendation*, *action recommendation*, and *object recognition*) separately, we found differences in the combination of cues that are statistically significant in predicting low certainty between scenarios (Table 14). Across all scenarios, *slower pace of speech* and *ending with a rising intonation* was statistically significant to differentiate

low certainty utterances from medium certainty. *Minimal emphasis on words* is statistically significant for the item recommendation and action recommendation scenarios, but not significant for the object recognition scenario. Using *lower volume* and *prolonged words* are only significant for the action recommendation scenario; and using *pauses* is only significant for the object recognition scenario.

Clustering analysis using gap statistic for low level of certainty across the measured cues as dimensions broken down by scenario type reveals **one** cluster in the data from the *item recommendation* scenario, **one** cluster in the data from the *action recommendation* scenario, and **two** clusters in the data from the *object recognition* scenario (Table 15). The low number of clusters for each scenario suggests that participants were similar to each other in using paralinguistic cues to express uncertainty.

Looking at the cluster centers, it seems like participants were using paralinguistic cues similarly between *items recommendation* and *action recommendation* scenarios, where they are using slightly lower volume, slower pace, no emphasis on words, some pauses, disfluencies and prolonged words, with ending some phrases in a rising intonation. *Object recognition* scenario differs slightly, where the majority of participants did not use as many pauses, disfluencies and prolonged words but ended more phrases with rising intonations (OR Cluster 1). On the other hand, some participants chose to use a slower pace with more pauses, disfluencies and prolonged words instead (OR Cluster 2).

In the high level of certainty section above, we discussed *item recommendation* and *action recommendation* scenarios are perceived to have **higher personal stake** as compared to the *object recognition* scenario, which could be a potential reason for the difference in using paralinguistic cues across interaction scenarios. Some participants specifically mentioned that the two recommendation scenarios felt similar to each other, while the object recognition scenario felt different from them:

“... option (item) recommendation and action recommendation, they felt almost the same. But there were some differences, (for) recognition.” [P92]; “I think on the options (recommendation scenarios), the differences were a lot more subtle.” [P99]

Comparing between high and low levels of certainty, there are fewer clusters overall as well as for each scenario in low certainty utterances, suggesting that there is more consensus among participants on how paralinguistic cues are being used to express low confidence. Also, we did not find any evidence that participants altered their expressions for low certainty utterances based the situation of the conversation (e.g. delivering a presentation vs. talking to a friend).

4 PHASE II - RECOGNIZING DIFFERENT LEVELS OF CERTAINTY

Phase I of our research provided an understanding of how people use and combine paralinguistic cues across different interaction contexts in order to express varying levels of certainty. While this brings us closer to understanding how voice-based agents can be design to use paralinguistic cues to express varying levels of certainty, given that what we learn about how humans employ these cues cannot necessarily be directly applied to agents as is, we also need to understand how recognition of certainty compares when paralinguistic cues are expressed by voice-based agents vs. when they are expressed by other people. Therefore, in Phase II of our research, we explored if low, medium, and high levels of confidence, expressed using the paralinguistic cues identified in Phase I, can be reliably recognized by people, and how people’s perceptions of these cues compare when they are uttered by agents vs. when they are uttered by other people.

In this section, we first describe our study design and data analysis approach, utilizing a survey to collect certainty ratings for phrases expressing different levels of certainty across interaction scenarios and communication partner

(human vs. agent). We then present our findings on the recognition of certainty overall, as well as drilling down to communication partner, gender, and scenario differences.

4.1 Study Design and Data Analysis

We designed a survey asking participants to rate the level of certainty expressed by a human or a voice-based agent for various audio recordings. These recordings were selected and modified from our Phase I study using the following selection procedure:

- (1) We used audio clips from female participants only. As most voice-based agents in the market (e.g. Amazon Alexa) are using female voices as their default setting, we wanted to compare the ratings between human vs. agent using the same gender.
- (2) We selected audio clips that match the paralinguistic cues at each level of certainty as outlined in Table 5.
- (3) These audio clips were then normalized for volume to eliminate arbitrary loudness differences between them.
- (4) We randomly selected two phrases for each scenario at each level of certainty, resulting in a total of 18 recordings selected to represent human-human communication ($3 \text{ scenarios} \times 3 \text{ levels of certainty} \times 2 \text{ phrases}$).
- (5) In order to create recordings to be used for human-agent communication, we used the speech-to-speech synthesis tool from Eleven Labs³ to convert each of the 18 selected recordings to the Serena voice, which is a pleasant American voice similar to popular voice agents such as Amazon Alexa or Google Assistant. Through this synthesis, we created 18 additional recordings preserving the paralinguistic clues from the original audio clips while creating the impression that these recordings were expressed by the same voice agent.

As such, we created 36 recordings with the following characteristics:

- Gender (1): female voices only
- Levels of certainty (3): high, medium and low
- Scenarios (3): item recommendation, action recommendation, and object recognition
- Phrase variation within each scenario (2)
- Communication partner (2): human vs. agent

These 36 recordings were then incorporated into a survey, where each question presented a dialog situation and an associated recording, and asked users to select a value between 0 (not confident at all) and 10 (very confidence) on the level of certainty expressed by the recording. We differentiated between communicating with a human vs. agent partner through the wording of the question, such as *"Imagine you are asking a friend .."* vs. *"Imagine you are asking a voice agent (e.g. Amazon Alexa) .."*.

Through Prolific⁴ we recruited 100 subjects from Canada and United States who are fluent in English to participate in the survey, with gender balancing of 50% female and 50% male. Each subject was presented with 18 questions, where they were randomly assigned to either the human or the agent recording for each question. The ordering of the questions in the survey was also randomized to avoid ordering bias. With the between-subject design of randomly assigning human vs. agent recordings, we collected on average 50 ratings per recording for analysis.

To analyze the ratings, we generated a summary statistics table with the average and standard deviation of the ratings for each level of certainty (Table 16a), as well as visualized their distributions as a box plot (Table 16b). We then applied pairwise comparison using the linear mixed-effect model [73] to understand the statistical significance between

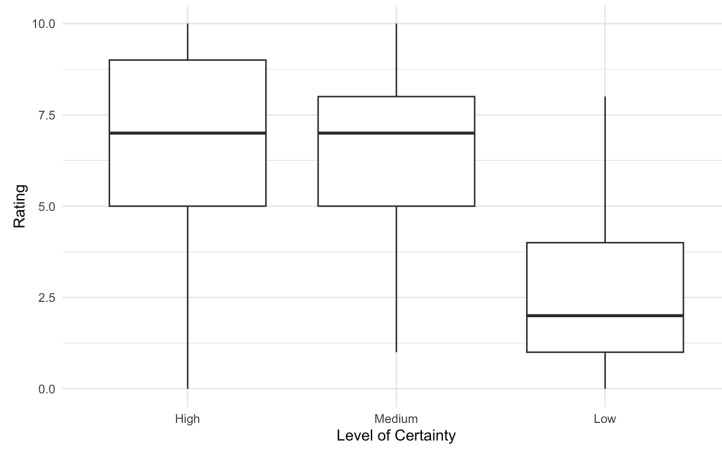
³<https://elevenlabs.io/speech-synthesis?task=speech-to-speech>

⁴<https://www.prolific.com/>

Table 16. Ratings for Each Level of Certainty

	Average	Stdev
High	6.56	2.54
Medium	6.63	2.29
Low	2.82	2.12

(a) Summary statistics table



(b) Boxplot visualization of ratings at each level of certainty

	High vs. Medium	Low vs. Medium	High vs. Low
p-value	0.8339	<0.0001	<0.0001
effect size	-0.033	1.855	1.822

(c) Pairwise comparison between levels of certainty

the recognition for each level of certainty (Table 16c). The linear mixed-effect model was chosen for our analysis to control for random effect variations within each participant while comparing the fixed effects between the certainty levels. We also calculated the standardized effect sizes using the R package `emmeans::eff_size` [79] to compare the mean differences between groups. Lastly, we drilled down across different dimensions of the data to understand if there are statistically significant differences in the recognition of certainty based on communication partners (human vs. agent) (Table 17), gender (male vs. female) (Table 18), and scenarios (item recommendation, action recommendation, and object recognition) (Table 19).

4.2 Findings

Our findings demonstrate that participants were able to distinguish low certainty expressions from high and medium levels of certainty, but cannot distinguish high certainty expressions from medium level of certainty. We also found recognition differences if the conversation partner is a human or an artificial agent, where agent utterances were consistently recognized as less certain as compared to human utterances using the same paralinguistic cues at all levels of certainty. Lastly, we found scenario differences, where action recommendation is recognized different from item recommendation and object recognition.

Looking at the summary statistics table (Table 16a), the rating for low certainty phrases were much lower at 2.82 as compared to high and medium levels of certainty, while there is only a small difference between the ratings for high (6.56) and medium (6.63). Surprisingly, the average for medium certainty phrases are slightly higher than high certainty phrases. Visualizing the spread of the ratings through boxplots (Table 16b), there is a wider range of participants' ratings for high certainty phrases (standard deviation 2.54) compared to medium (2.29) and low phrases (2.12). Pairwise

comparison showed statistical significance for low level of certainty compared to medium and high levels with large effect sizes (1.855 and 1.822 respectively), but did not reveal differences between high and medium (Table 16c). Based on these findings, we extrapolate that the paralinguistic cues we identified in Phase I (Table 5) can be reliably used to express low level of certainty that can be easily recognized by the conversation partner. However, the cues associated with high level of certainty, while distinguishable from low certainty phrases, cannot be easily differentiated from medium level of certainty.

Table 17. Human vs. Agent Rating Comparison

	High	Medium	Low
Human	7.33	7.03	3.11
Agent	5.83	6.26	2.53
Diff	1.50	0.77	0.58

(a) Average rating for each partner at each level of certainty

		High vs. Medium	Low vs. Medium	High vs. Low
Human	p-value	0.0855	<0.0001	<0.0001
	effect size	0.181	2.064	2.244
Agent	p-value	0.0872	<0.0001	<0.0001
	effect size	-0.172	1.887	1.715

(b) Pairwise comparison between levels of certainty for each partner

		High	Medium	Low
Human vs. Agent	p-value	<0.0001	<0.0001	0.0033
	effect size	-0.620	-0.366	-0.256

(c) Pairwise comparison between partners at each levels of certainty

4.2.1 Human vs. Agent Comparison.

Next, we wanted to understand if there are differences in certainty recognition when conversing with a human vs. an artificial agent partner using the same paralinguistic cues to express levels of certainty. Comparing between levels of certainty within each conversation partner, the recognition of certainty is similar to overall findings, where the difference is not significant between high and medium, but it is significant between low and medium as well as high and low with large effect sizes (Table 17b). Comparing the ratings between humans and agents, we noticed that the utterances in scenarios with artificial agents are consistently rated lower at all levels of certainty (Table 17a). The most significant difference appeared in the rating for high level of certainty, where utterances in scenarios with human conversation partners were rated on average at 7.33 while agents were rated on average at 5.83. Pairwise comparison showed that the difference between human and agent utterances is significant at all levels of certainty (Table 17c). In terms of the magnitude of the differences between levels, high level of certainty has a moderate effect of -0.62, whereas medium and low levels of certainty have a low effect of -0.366 and -0.256 respectively. This indicates that while agents are consistently rated lower in certainty than humans across all levels of certainty, the rating difference between human and agent is greater for high certainty utterances.

4.2.2 Listener Gender Comparison.

Looking at listener-gender differences in the recognition of certainty, the average ratings for each level of certainty are similar between male and female participants (Table 18a). Within each conversation partner, the recognition of certainty is similar to overall findings, where the difference is not significant between high and medium, but it is significant between low and medium as well as high and low with large effect sizes (Table 18b). Pairwise comparison between genders revealed no statistical significance with negligible effect sizes at each level of certainty (Table 18c). This demonstrates that listener gender differences do not influence recognizing certainty levels associated with paralinguistic cues.

Table 18. Listener Gender Rating Comparison

	High	Medium	Low
Male	6.49	6.51	2.92
Female	6.63	6.74	2.71
Diff	-0.14	-0.23	0.21

(a) Average rating for each gender at each level of certainty

		High vs. Medium	Low vs. Medium	High vs. Low
Male	p-value	0.9933	<0.0001	<0.0001
	effect size	-0.009	1.636	1.627
Female	p-value	0.7314	<0.0001	<0.0001
	effect size	-0.062	2.126	2.064

(b) Pairwise comparison between levels of certainty for each gender

		High	Medium	Low
Male vs. Female	p-value	0.6964	0.4569	0.5347
	effect size	0.0529	0.105	-0.128

(c) Pairwise comparison between gender at each levels of certainty

Table 19. Scenario Rating Comparison (IR = Item Recommendation; AR = Action Recommendation; OR = Object Recognition)

	High	Medium	Low
IR	7.09	6.27	2.58
AR	5.25	7.06	2.96
OR	7.32	6.54	2.91

(a) Average rating for each scenario at each level of certainty

		High vs. Medium	Low vs. Medium	High vs. Low
IR	p-value	<0.0001	<0.0001	<0.0001
	effect size	0.468	2.100	2.567
AR	p-value	<0.0001	<0.0001	<0.0001
	effect size	-0.911	2.060	1.150
OR	p-value	0.0005	<0.0001	<0.0001
	effect size	0.379	1.748	2.127

(b) Pairwise comparison between levels of certainty for each scenario

		High	Medium	Low
IR vs. AR	p-value	<0.0001	0.0002	0.0352
	effect size	-0.919	0.403	0.247
IR vs. OR	p-value	0.4738	0.3583	0.0846
	effect size	-0.116	-0.136	-0.212
AR vs. OR	p-value	<0.0001	0.0204	0.9328
	effect size	-1.035	0.267	0.035

(c) Pairwise comparison between scenarios for each level of certainty

4.2.3 Scenario Comparison.

We discovered that participants recognized certainty differently between the three scenarios *item recommendation* (IR), *action recommendation* (AR), and *object recognition* (OR). While for the *item recommendation* and *object recognition* scenarios the levels of certainty were recognized in the correct relative order, this was not the case for the *action recommendation* scenario, where *medium* certainty was interpreted higher (average = 7.06) than *high* certainty itself (average = 5.25) (Table 19a). Pairwise comparison between levels of certainty for each of the scenarios showed statistical significance between all pairings (Table 19b). The effect sizes of the differences between low vs. medium and low vs. high are large for all scenarios, while the effect sizes between high vs. medium are moderate for the item recommendation and object recognition scenarios, and large for the action recommendation scenario. This finding suggests that participants can differentiate between levels of certainty for all scenarios, but rate high and medium certainty phrases consistently wrong for the action recommendation scenario.

Comparing the ratings between scenarios at each level of certainty confirms the finding that *action recommendation* scenario phrases are recognized differently for high and medium levels of certainty, as we see statistical significance between action recommendation vs. item recommendation, as well as action recommendation vs. object recognition scenarios at high and medium levels of certainty (Table 19c). The effect size of the difference between the action recommendation scenario compared to other scenarios for high level of certainty is large (effect size -0.919 for IR vs AR; -1.035 for AR vs. OR), while the effect size is moderate for medium level of certainty (effect size 0.403 for IR vs. AR; 0.267 for AR vs. OR). This implies that the action recommendation scenario differs from other scenarios in that high and medium confidence are recognized differently, with a more pronounced difference for high certainty expressions.

5 DISCUSSION

The combination of findings from Phase I and II reveal that it is easier to express and recognize low certainty than it is to express or recognize high certainty. This is indicated by our Phase I findings which illustrate that there are more variations in how people use and combine paralinguistic cues to express high certainty compared to low certainty, suggesting that there is less agreement in how to express high certainty compared to low certainty. Furthermore, in Phase II, participants were more consistently able to recognize low-certainty phrases than high-certainty phrases, suggesting that it is easier to recognize low certainty both when it comes from voice-based agents and other people. Although this is the case, Phase II findings also revealed that people tend to rank statements uttered by voice-based agents as lower confidence than those uttered by other people, no matter the certainty level that the interlocutor is trying to convey, and even when the same paralinguistic cues are employed; thus illustrating that certainty expressions are perceived differently in human-agent interactions vs. in human-human interactions.

In this section, we discuss the factors that may be impacting the expression and recognition of certainty in human-human vs. human-agent communication overall and across interaction contexts, outline a set of design guidelines for voice-based agents using paralinguistic cues to communicate certainty, and discuss ethical considerations such as malicious manipulation and reinforcement of existing stereotypes.

5.1 Recognition of Certainty in Human-Human vs. Human-Agent Communication

Given that there are differences in how certainty is perceived and recognized in human-agent interactions vs. in human-human interactions, voice-based agents need to employ paralinguistic cues differently from how humans employ such cues.

One of the reasons for this could be the perceived relationship difference for communications between human-human vs. human-agent. Based on previous literature, we know that most people perceive human-agent interaction as a transactional relationship to achieve a functional goal, without the desire to building deeper relationships with artificial agents [20]. Also, there is the perceived power imbalance that agents are considered to be user-controlled tools rather than an entity with equal social status [20, 29]. With these factors of transactional relationship and unequal status, it is possible that users unconsciously assign less confidence to the recommendations that agents provide, resulting in the systematic difference in the lower certainty ratings for agent utterances.

Another reason for the difference in certainty ratings between human-human and human-agent communication could be that users do not feel that human-like expressions such as paralinguistic cues are relevant to agents. Kim et al. [65] found that pragmatic markers are generally perceived negatively by users as too extraneous and unnecessary in human-agent dialogue, but acceptable in human-human conversations. Based on the existing voice-based virtual assistants in the market (e.g. Alexa, Siri), users perceive these agents as flat or mechanical sounding without conveying

any emotions in their utterances [29]. This may explain why phrases with high or medium levels of certainty are perceived as lower confidence for agents as compared to human interactions. As for the difference in ratings for phrases with low certainty, users generally expect agents to provide transactional-based factual information that is usually not compatible with low level of confidence [20]. Jeong et al. [61] showed that conversational agents using fillers in their utterances are perceived as less intelligent in task-oriented situations. These factors may lead to lower perceived confidence ratings for agents using paralinguistic cues to convey low levels of confidence.

5.2 Differences Across Contexts and Between Different Levels of Certainty

The fact that there are multiple combinations of values along the same cues that can be used to convey low and high certainty, is important and useful for if the designer needs to adjust the agent’s voice to adapt to different contexts where it cannot be as flexible in adjusting certain paralinguistic cues (e.g., different cultural or situational contexts where the agent is expected to always speak slowly or quietly, or contexts in which the agent is speaking to a non-native speaker and needs to slow down its speech to be better understood). In such contexts, certain cues must always remain constant. Our findings illustrate that even if certain cues must remain constant, perceptions of low and high confidence can often still be achieved by adjusting alternative cues, depending on the scenario at hand.

Furthermore, our findings suggest that while there is a clear set of paralinguistic cues that can be used to reliably express uncertainty across a variety of interaction contexts, the same cannot be said for expressing high certainty. The paralinguistic cues that people choose to express *high confidence*, as well as how these cues are perceived depend on the *context of the communication exchange*, such as the interaction scenario, the social setting of the conversation, and the familiarity between the interlocutors.

5.2.1 Interaction Scenario. We found that the expression and recognition of high certainty is dependent on the scenario at hand. One of the key factors differentiating the scenario types we investigated is the level of *personal stake* involved (i.e., the potential severity of the consequences of ‘getting it wrong’). In our Phase I study, participants chose to use pause and emphasis in different ways to convey high confidence in situations that may be perceived to be higher risk, such as item recommendation and action recommendation; whereas for object recognition (which may be perceived as lower risk) participants did not use pauses or emphasis to convey high confidence. Previous research has suggested that factors such as one’s *desire to persuade* (e.g., [89, 90]), *susceptibility to social pressure* (e.g., [22, 23, 42, 43, 90]), and the *potential severity of the consequences of one’s decision* (e.g., [24, 39, 90]) all of which can differ depending on the scenario at hand, may affect displays of certainty. In the *item recommendation* and *action recommendation* scenarios, the goal is to persuade someone to pick an option, and the choice that the person makes has consequences, some of which could potentially lead to social acceptance or ridicule. Thus, these factors of *desire to persuade*, *susceptibility to social pressure*, and the *potential severity of the consequences of one’s decision* might also be affecting participants’ expressions and recognition of certainty in *item recommendation* and *action recommendation* scenarios.

5.2.2 Social Setting. The social setting of the conversation is another communication context that could affect the use of paralinguistic cues for high certainty. In Phase I, participants mentioned that they express high confidence differently depending on the situation, such as conversation with friends or making a presentation at work. Specifically, they mentioned that there are social norms associated with the use of paralinguistic cues, such as not using any conversational fillers at work as it may be seen as casual and unprofessional.

5.2.3 Familiarity Between Interlocutors. The familiarity between interlocutors is a communication context that can affect how levels of certainty are perceived. In Phase I, participants mentioned that they considered medium certainty as the natural baseline on how they would normally speak, and varied high and low certainty based on medium. However, this requires the communication partner to be familiar with the speaker’s natural way of speaking in order to deduce variations in their speech. For example, for someone who expresses themselves with higher confidence for their everyday speech, people get used to hearing the types of cues used to express high certainty and begin to associate those cues with a medium or default baseline for that individual. This also has implications for the design of voice assistants meant to be used over a longer period of time, because as people get used to interacting with such agents over a longer term, they could begin to associate their normal speaking cues with medium certainty.

5.3 Design Recommendations

From our findings, we present the following design recommendations for the expression of certainty and uncertainty by voice-based artificial agents:

Design Recommendation 1: *Use familiar voices and speaking patterns to express medium certainty.*

Our findings suggest that users associate one’s normal speaking cues with medium certainty, which they perceive as conveying neutrality (section 3.2.1). Existing research shows that familiarity with the speaker is a key factor for listeners to recognize their speaking cues and to correctly identify their identity [53, 64, 76]. When listeners are more familiar with a speaker’s voice, they are more likely to correctly interpret their cues [104], which could include their ‘default baseline’ voice that the listener normally hears. Thus we recommend that designers use how agents speak today (e.g., the types of voices and cues associated with the normal speech of voice assistants like Alexa, Google Assistant, and Siri) as a familiar baseline for agents to express medium certainty.

Example: *(use default agent voice with no modifications)* I would suggest going with this option.

Design Recommendation 2: *To express low certainty consistently across a variety of interaction contexts, slow down the agent’s pace, avoid emphasis of words, drag out some words, and end phrases with rising intonation.*

For expressing low certainty, our findings suggest that designers should modify the agent’s default voice to slow down its pace (by about 23%), not emphasize any words, drag out some words (about 1.1 words per phrase), and end the phrase on a rising intonation (Table 11). We found these cues are universally associated with low certainty across cultural differences (e.g. native vs. non-native speaker) (section 3.2.3.1) and interaction scenarios (section 3.2.3.3). Also, phrases using these cues can be reliably recognized as low confidence as compared to high and medium certainty utterances (section 4.2).

While we outlined the specific variations of the cues (e.g. slow down by 23%) to convey low certainty, we found our participants use these cues interchangeably in their expressions (section 3.2.3.2). Therefore, it is possible that the existence of these cues are enough to trigger the recognition of uncertainty without following the specifics of how to implement them.

Additionally, based on our scenario analysis (section 3.2.3.3), the agent can use some pauses and disfluencies in its expressions for low personal stake scenarios such as non-professional and casual contexts.

Example: *(slow down by 23%) Umm .. I would (prolong word)_suggest going with this (rising intonation)_option?*

Design Recommendation 3: Reserve expressions of high certainty for rare, more critical situations, where the agent needs to urgently persuade or convince the user.

We recommend that agents express high certainty *only in exceptional instances* where the user has a substantial personal stake and gaining the user’s confidence in the agent’s stance is critical (for example, where the agent needs to persuade the user on a time-sensitive matter). In other instances, where gaining the user’s trust or confidence is not the highest concern, having the agent express medium certainty and letting the user exercise a healthy level of skepticism of the agent’s conclusion may be acceptable, especially so as to prevent the user from over-relying on the agent and its statements.

This recommendation is based on our findings that agents tend to be perceived as expressing lower confidence compared to humans using the same paralinguistic cues across interaction contexts, therefore it is important to reserve the use of high certainty expressions for critical situations (section 4.2.1). Also, the paralinguistic cues used to express high certainty are not consistent across scenarios (section 3.2.2.3). One potential reason for this inconsistency is the perceived personal stake of the scenario, where users felt they needed to convey more confidence when the perceived personal stake is higher. Cultural difference may also be a contributing factor, as we found differences in expressions between native vs. non-native speakers in the use of pauses and prolonged words (section 3.2.2.1).

Therefore, in exceptional circumstances where agents need to express high certainty, our findings suggest that the agent can raise its speaking volume (by about 5%), increase its speaking pace (by about 13%), add more emphasis (e.g., emphasizing one word per phrase), and end phrases with a falling intonation to express higher confidence than normal (Table 6).

Example: (*louder by 5%, speed up by 13%*) I would suggest going with (*emphasis*)_this (*falling intonation*)_option.

5.4 Ethical Considerations

While the intention of the paralinguistic cues identified in our findings is to calibrate trust between users and agents by expressing the level of certainty according to confidence levels in the algorithm, it has the potential to be used for malicious intent to influence user behaviours. Existing research shows that manipulating the voice of an agent such as making it sound more familiar can increase the agent’s perceived credibility and likeability, persuading users to be more likely to comply with its requests [17]. Similarly, designers can leverage paralinguistic cues to express a high level of confidence to users that is not aligned with the actual confidence level of the underlying algorithm in order to mislead users to believe the agent is more competent than its capabilities.

Another ethical consideration is the rich set of information about the speaker’s characteristics that are conveyed by paralinguistic cues in human-human communication, including gender, age, socioeconomic and cultural background [116]. These cues can also reflect of the emotional state of the speaker as well as reflect the relationship closeness between the two interlocutors [122]. While our findings demonstrated the mechanics of the identified cues that are associated with different levels of confidence, we do not currently have a comprehensive understanding on all potential perceptions that may be elicited through these cues. Therefore, it is possible that agents using paralinguistic cues to express different levels of certainty may elicit additional perceptions from their users leading to unintended consequences.

Lastly, paralinguistic cues associated with different levels certainty may be reflecting existing societal stereotypes. In our study, we found that participants believe that lower pitch is associated with high level of confidence, while higher pitch is associated with low level of confidence. Some female participants even reflected that they consciously lowered

their voice to “sound more like a man”. Lower pitch is also found to be perceived as more authoritative and persuasive [31, 124]. These perceptions of voices with lower pitch may be demonstrating users’ unconscious bias to associate higher level of confidence and authority with a male-sounding voice instead of a higher pitch voice that is commonly associated as female-sounding. Therefore, it is important for designers to be cautious when utilizing paralinguistic cues in agents to make sure that they are not reinforcing existing stereotypes.

5.5 Limitations and Future Directions

In this section, we outline the limitations and methodological considerations of our work, and suggest directions for future research to address these limitations.

More research is needed to further understand the underlying factors driving the differences in certainty perceptions between human-human and human-agent communication. These differences in certainty ratings combined with existing body of knowledge suggest that there are fundamental differences in how users perceive paralinguistic cues between human-human and human-agent communications, and thus we cannot simply translate the cues from human interactions to be implemented in agents.

In Phase I of our study, we designed a controlled experiment where we asked participants to express pre-determined phrases associated with each scenario by varying their paralinguistic cues. While this approach allowed us to separate non-verbal cues from verbal content for analysis, we may be missing expressions that are being unconsciously used in more natural settings where participants have freedom to choose phrases that are more aligned with their everyday speech.

Secondly, for Phase II of the study to recognize the levels of certainty, we were not able to preserve the loudness aspect of the non-verbal cues in each phrase due to the systematic volume differences in participant recordings influenced by their study environment and their computer equipment. As such, we do not know whether loudness variation would have affected recognition of levels of certainty.

Third, for Phase II, we aimed to keep the voices consistent in most qualities, except the paralinguistic cues being tested. Therefore we decided to keep the gender of the voices consistent, and since most voice assistants on the market default to female voices, we chose to test with only female voices to test with voices that are consistent with most users’ current experiences with voice assistants. However, we recognize that there are limitations of testing only female voices, including that not all of the findings might extend to male voices given people’s tendencies to perceive male and female voices differently [13]. Additionally, this could present a risk of perpetuating existing stereotypes [91].

Fourth, it is unclear whether our findings can be generalized to apply to other cultures or languages outside of North America. The participants for all phases of the study are recruited from Canada and United States, who are fluent in standard American English. Also, while we selected our participants to include a diverse range of demographic background (e.g. age, native vs. non-native speakers), there may be other factors impacting the use of paralinguistic cues we have not captured in our studies, such as the impact of individual characteristics (e.g. level of empathy).

Therefore, we recommend further studies to address these limitations and to enhance our understanding on the association between paralinguistic cues and expressions of certainty, including studying within natural settings for expression of non-verbal cues, incorporating loudness cues for certainty recognition, testing a variety of voice genders including male, female, non-binary, and gender-neutral voices, expanding the study to other languages and cultures, as well as exploring the impact of individual characteristics.

Lastly, for future direction we should run an empirical study with an agent incorporating our design recommendations to express different levels of certainty using paralinguistic cues. This will allow us to understand whether these cues

can be appropriately implemented within an agent. Also, it will allow us to observe the interactions between the agent and users to see if these cues are sufficient to differentiate between the levels of certainty.

6 CONCLUSION

In this work, we share our findings of communicating different levels of certainty using paralinguistic cues through conversational interfaces for trust calibration. We found that seven paralinguistic cues are associated with the expression of levels of certainty conversationally: volume, pace, emphasis, disfluency, pause, pronunciation, and intonation/pitch (RQ1). We found that it is easier to express and recognize low certainty than it is to express and recognize high certainty, as there are more variations in how people express high certainty compared to low certainty, which suggests that there is less agreement among individuals on how to properly express high certainty. In a follow-up study, we found that people can consistently recognize low-certainty phrases, but had trouble distinguishing phrases between high- and medium-certainty for both human-human (RQ2.1) and human-agent (RQ2.2) conversations. Also, we uncovered that the recognition of certainty for human utterances were consistently higher compared to agent utterances, demonstrating a systematic difference between human-human vs. human-agent communication.

The findings above on the association between paralinguistic cues and different levels of certainty can be used by designers to communicate high and low certainty conversationally for trust calibration purposes. However, paralinguistic cues should be used in conjunction with the content of the communication to effectively convey the intended certainty level. Our analysis revealed that more research is needed to understand the influence of different scenarios, native vs. non-native English speakers, as well as the differences in certainty perceptions between human-human vs. human-agent. We urge designers to incorporate ethical perspectives into their considerations when using paralinguistic cues, including the potential for manipulation, eliciting unintended perceptions leading to unexpected consequences, and perpetrating existing stereotypes.

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